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UW24Subsys_About.1.2



[About the UltraWave24 Subsystem : Features](#)

Features

The UltraWave24 subsystem has the following features and capabilities:

- 6GHz source and measure
- Very low phase noise synthesizers and sources
- CW and modulated sources, modulated source bandwidth 160 MHz
- Multitone source with CW or modulated individual tones
- Receivers with programmable IF filters, including bypass, and two types of digitizers:
 - VHF (5 MHz-165 MHz bandwidth, 12 bit)
 - High speed (500 kHz-45 MHz bandwidth, 16 bit)
- Reference synthesizer providing eight DUT clock drive pins
- Universal ports and channels, so each 6 GHz MW instrument channel with 1:3 multiplexer has the same features and performance:
 - Independent source level control with level calibration using stabilized reference
 - 1 dB DSA to maximize source signal integrity and optimize receive SFDR
 - Pin diode switches for rapid settling, high isolation, and low loss
 - Wideband noise source for calibrated ENR
 - Receive path with low-noise amplifier
- Scalable architecture for high pin count devices and high multisite parallel efficiency

- Configurations ranging from basic minimum to maximum in incremental steps that add measure boards and synthesizer boards:

- Minimum: 24 universal ports, with four synthesizers (including the DUT reference) and four receivers
- Full: 96 universal ports, with eight synthesizers (including two Reference sources) and 16 receivers

- High parallel capabilities, such as:

- A configuration with 72 ports in four test head slots (three measure boards and one synthesizer board with four synthesizers with reference) provides up to 24 parallel source ports or 12 parallel measure ports (maximum of 24 active paths)
- Background data handling and DSP

- SyncLink synchronization with pattern control and pattern sets

- Software “logical instrument” interfaces to the MW hardware functions that simplify programming of subsystem resources for measurements

- Support for LTE-Advanced and 802.11ac

- DIB compatible with UltraWave 12G

For hardware specifications, see [UltraWave24 Specifications](#).



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[About the UltraWave24 Subsystem](#) : Safety Information

Safety Information

When operating instruments using the UltraWave24 subsystem, always follow general UltraFLEX test system safety guidelines. See [About Operator and Safety Information](#).

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About the UltraWave24 Subsystem : Key Terms

Key Terms

add path	Signal combiner allowing two source signals to be added to create either a two-tone signal or a multitone signal if one or both signals are from a modulated source.
board set	A UltraWave24 instrument requires a minimum of two boards: a synthesizer board and a measure board. Larger configurations have more of each kind. See Supported UltraWave24 Configurations.
CW	Continuous wave. A signal type equivalent to sine wave.
DUT clock channels	Sets of eight clock signals on a MW Synthesizer Reference board available at the DIB interface. Each channel has an independent disconnect relay.
IF	Intermediate frequency. Frequency at the output of receiver downconverter.
LO	Local oscillator. Used with a mixer to either modulate a carrier or down convert signals.
logical instrument	A set of MW functions configured and connected to perform as a MW instrument with specific capabilities. Examples are MWSource and MWReceiver. Each logical instrument has both VBT language and GUI debugging tool programming interfaces.

MW instrument channel or MW channel	An individual low-noise, full-bandwidth signal path in the UltraWave24 subsystem available to source and measure functions. Each MW channel is “universal” - all have the same features and performance. Contained in a single front-end module on a MW Measure board.
mux	Multiplexer. One-to-many switch allowing a MW instrument channel to be connected to three RF I/O ports (tester channels). MW multiplexers are bidirectional.
MW	Microwave
physical instrument	A set of MW hardware functions configured and connected to perform as a MW logical instrument.
port	An R/F I/O signal connector at the DIB interface providing a signal path to either the DUT or a DIB-mounted application-specific circuit. Each RF port corresponds to a MW tester channel. Compare to a spring-loaded pin.
RF	Radio frequency
SMPM connector	Blind-mate connector type for each RF I/O port DIB connection.
splitter	Network that splits power across two or more outputs. Allows a single source to drive multiple loads.
VCA	Voltage-controlled attenuator or voltage-controlled amplifier.

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About the UltraWave24 Subsystem : UltraWave24 Software Licensing Requirements

UltraWave24 Software Licensing Requirements

UltraWave24 instruments require one or more licenses to use the following features:

- Source Board (up to two per test system)
 - UltraWave24 Modulated Source Bandwidth
 - 2 Mod Source Per Source Board
 - UltraWave24 Dual Synthesizer
 - UltraWave24 Dual AWG
 - UltraWave24 Source Frequency >700 MHz
- Measure Board (four per test system)
 - UltraWave24 Universal Port
 - UltraWave24 Receiver Bandwidth
 - UltraWave24 Noise Source

–	UltraWave24 1 Port VNA
–	UltraWave24 Receiver Frequency >700 MHz
–	UltraWave24 Receiver Enable
–	UltraWave24 Enhanced Measurement
–	UltraWave24 Standard Measurement
–	UltraWave24 Standard Multitone
–	UltraWave24 Enhanced Multitone

The table below lists licensed features for the UltraWave24.

UltraWave24 Licensed Features

Feature Name	Feature	Comments
UW24_1PortVNA1	1 Port VNA Option	Counted per measurement board. Allows for 1-port VNA measurement on any port on the MWMeasure Board.
UW24_AWG	Modulated Source, 80 MHz and 160 MHz Bandwidths	Counted license. One license required per modulated source. Can be enabled at 80 MHz and 160 MHz. Two modulated sources allowed per source board.
UW24_DualAWG1	Dual AWG	Counted license. Two modulated sources allowed per source board.
UW24_DualSynthesizer	Dual Synthesizer	Counted license. Enables second source per source board.
UW24_DutRef1	DUT Reference	Counted license. One license required per DUT reference. One

DUT reference allowed per source board.

UW24_EnMeas1

Enhanced
Measurement

Counted license. Up to four measure instruments can be connected per measure board.

UW24_EnMultiTone1

Enhanced Multitone

Counted license. Up to four measure instruments can be connected per measure board.

UW24_NoiseSource1

Noise Source

Counted license. One license required per measurement board. One noise source allowed per measure board.

UW24_Ports1

RF Port

Counted license. One license required per port. Each license enables 24 ports on a single RF measurement board. Up to 96 ports per test system.

UW24_RcvEnable1

Receiver Enable

Counted license. One license required per measurement board. Up to four receivers per measure board.

UW24_RcvFrequency700

Receiver Frequency
> 700 MHz

Counted license. One license required per receiver operating above 700 MHz. One allowed per measure board.

UW24_Receivers

Receiver

Counted license. One license required per receiver operating at 45 MHz. Can be enabled at 100 MHz and 160 MHz. Up to four receivers per measure board.

UW24_SourceFrequency700

Source Frequency >
700 MHz

Counted license. One license required per source operating above 700 MHz. One allowed per source board.

UW24_StdMeas1	Standard Measurement	Counted license. One license required per UltraWave24 measure board. Up to two measure instruments can be connected per measure board.
UW24_StdMultiTone1	Standard Multitone	Counted license. One license required per UltraWave24 measure board. Up to two multitone instruments can be connected per measure board.

In an offline simulated or current configuration file, these licenses appear as follows:

<LICENSE_ENABLERS>

#Off-line enabling levels

#FEATURE	ENABLED	LEVEL	LICENSE COUNT	LEGAL ENABLING POINTS
UW24_AWG	160	4	160	
UW24_DutRef1	1	3	1,0	
UW24_Ports1	1	96	1,0	
UW24_Receivers	160	16	160	
UW24_NoiseSource1	1	4	1,0	
UW24_1PortVNA1	1	4	1,0	

</LICENSE_ENABLERS>

For more information, see [IG-XL Licensing](#) in the IG-XL Administration section.

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UltraWave24 Examples

UltraWave24 Examples

This section contains IG-XL workbook examples showing all IG-XL data and VBT code necessary for a complete running program:

- | | |
|---|---|
| • | MWSource-MWReceiver Loopback Test |
| • | MWModulatedSource-MWReceiver AM Signal Test |

For more information, see:

- | | |
|---|---|
| • | UltraWave24 Instrument Programming |
| • | Individual MW instruments in the VBT Syntax Reference, such as MWSource |



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UltraWave24 Examples : MWSource-MWReceiver Loopback Test

MWSource-MWReceiver Loopback Test

The MWSource-MWReceiver loopback test verifies the connection from the MWSource to the MWReceiver through a cable and tests the amplitude of a sourced RF signal. The test program tests two sites.

A MWSource generates CW signals on two ports. Two MWReceivers downconvert the two signals and digitize them. (A site loop is not required because there is a receiver and digitizer for each site.) The amplitudes are calculated using a DFT DSP method.

The program runs both online and offline. Offline, a function in the DSP module generates a received waveform for the DFT routine. Since the program is dual site, two cables are required to run online.

The MW Measure HD board is installed in test head slot 2. A MW Synthesizer HD board must be internally connected to the MW Measure HD board, but the test head slot does not affect the channel map.

This workbook example contains the following components:

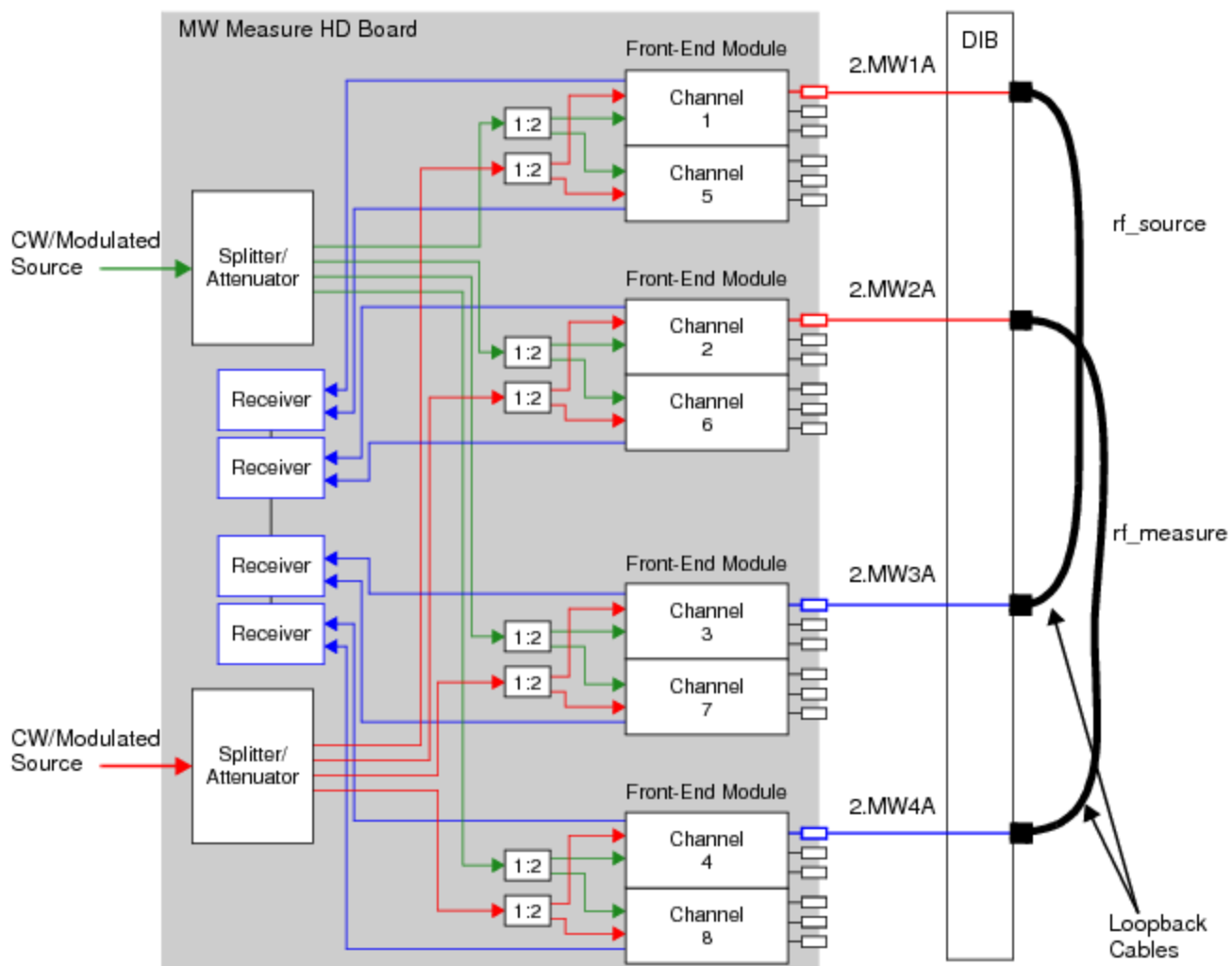
- | | |
|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBt Code |
| • | DSP Code |

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UltraWave24 Examples : MWSource-MWReceiver Loopback Test : DUT Connections

DUT Connections



UltraWave24 MWSource-MWReceiver Loopback Test Connections

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UltraWave24 Examples : [MWSource-MWReceiver Loopback Test](#) : DataTool Sheets

DataTool Sheets

This workbook example includes the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet
- Mixed Signal Timing Sheet
- Waveform Definition Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	rf_source	Analog
	rf_measure	Analog

Channel Map Sheet

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
rf_source		MW	2.MW1A	2.MW2A
rf_measure		MW	2.MW3A	2.MW4A

Flow Table Sheet

Flow Table

					Flow Domain:													
		Gate			Command				Limits		Datalog Display			Bin Number		Sort Number		
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Scale	Units	Format	Pass	Fail	Pass	Fail	Result
					Test	loopback_cw												
					Use-Limit	loopback_cw	rf_pow	99	-40	20				1	99	1	99	Fail

Test Instances Sheet

Test Instances

Test Procedure			
Test Name	Type	Name	Called As
loopback_cw	VBT	RF_meas_loopback	

The test instance named loopback_cw has the following test-specific parameters and values as shown in the Instance Editor:

Instance Editor [VBT Test 'RF_meas_loopback']

Instance:

Parameters Timing and Levels Limits

Name	Value
SrcPin	rf_source
RecvPin	rf_measure
RF_Freq	900000000
RF_power	-20
IF_freq	2512500
capture_FS	12500000
capture_size	1000

Mixed Signal Timing Sheet

The MW instrument does not use the Mixed Signal Timing sheet.

Waveform Definition Sheet

The MW instrument does not require a Waveform Definition sheet.

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[UltraWave24 Examples : MWSource-MWReceiver Loopback Test](#) : VBT Code

VBT Code

```
Public Function RF_meas_loopback(SrcPin As PinList, RecvPin As PinList, _  
    RF_Freq As Double, RF_power As Double, _  
    IF_freq As Double, capture_FS As Double, _  
    capture_size As Long) As Long
```

```
Dim RfSignal As String
```

```
Dim RfCapturedWave As New DSPWave
```

```
Dim measured_power As New SiteDouble
```

```
On Error GoTo errHandler
```

```
RfSignal = "RfCapture"
```

```
With TheHdw.MWSource(SrcPin)
```

```
    .amplitude = RF_power
```

```
    .Frequency = RF_Freq
```

```
End With
```

```
With TheHdw.MWReceiver(RecvPin)
```

```
    .Signals.Add RfSignal
```

```
    If TheExec.TesterMode = testModeOnline Then
```

```
        RfCapturedWave = .Signals(RfSignal).DSPWave.Pins(RecvPin)
```

```
    End If
```

```
End With
```

```
With TheHdw.MWReceiver(RecvPin)
```

```
    .AmplitudeExpected = RF_power
```

```
.CenterFrequency = RF_Freq
.IFFrequency = IF_freq
.capture.SampleRate = capture_FS
.capture.SampleSize = capture_size
.GainSetup = tlmwGainSetupNormal
End With

TheHdw.MWSource(SrcPin).Connect
With TheHdw.MWReceiver(RecvPin)
    .Connect
    .Signals(RfSignal).Trigger
    .WaitForCaptureDone
    .Disconnect
End With
TheHdw.MWSource(SrcPin).Disconnect

TheHdw.DSP.ExecutionMode = tlDSPModeHostDebug

Dim CurrSite As Variant
If TheExec.TesterMode = testModeOffline Then
    For Each CurrSite In TheExec.Sites
        Call rundsp.create_offline_capture(RF_power, IF_freq, capture_FS, _
            capture_size, RfCapturedWave)
    Next CurrSite
End If

rundsp.measRF_power RfCapturedWave, IF_freq, measured_power
Call TheExec.Flow.TestLimit(measured_power, Unit:=unitCustom, _
    customunit:="dBm", forceresults:=tlForceFlow, ScaleType:=scaleNoScaling)

RF_meas_loopback = TL_SUCCESS

Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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UltraWave24 Examples : [MWSource-MWReceiver Loopback Test](#) : DSP Code

DSP Code

```
Public Function measRF_power(ByVal dsp_in As DSPWave, _
    ByVal IF_freq As Double, _
    ByRef measured_power_dBm As Double) As Long

    Dim bin As Long, phase_ignore As Double
    Dim ampl As Double
    bin = IF_freq / dsp_in.FrequencyResolution
    ampl = dsp_in.CalcAmplitudeFromTimeDomain(bin, phase_ignore)
    measured_power_dBm = (10# / Log(10#)) * Log(1000# * ampl * ampl _
        / (2 * 50#))

    measRF_power = TL_SUCCESS
End Function

Public Function create_offline_capture(ByVal power_dBm As Double, _
    ByVal IF_freq As Double, _
    ByVal capture_rate As Double, _
    ByVal sample_size As Long, _
    ByRef capture As DSPWave) As Long

    Dim omega As Double
    Dim ampl As Double

    ampl = 10# ^ ((power_dBm - 10#) / 20#)
    omega = 2 * Pi * IF_freq / capture_rate
```

```
Call capture.CreateSin(omega, 0, sample_size)
capture = capture.Multiply(ampl)
capture.SampleRate = capture_rate
create_offline_capture = TL_SUCCESS
End Function
```

UW24Subsys_Apps.6.07

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UltraWave24 Examples : MWModulatedSource-MWReceiver AM Signal Test

MWModulatedSource-MWReceiver AM Signal Test

The MWModulatedSource-MWReceiver AM Signal test verifies the connection from the MWModulatedSource to the MWReceiver through a cable and tests the power in the sidebands of an amplitude modulated RF signal. The test program tests two sites.

The MWModulatedSource generates an AM signal using IF signal data. The signal is connected at two ports. Two MWReceivers downconvert the signal and digitize it. (A site loop is not required because there is a receiver and digitizer for each site.) The sideband amplitudes are calculated using DFT DSP methods.

The program runs both online and offline. Offline, a function in the DSP module generates a received waveform for the DFT routine. Since the program is dual site, two cables are required to run online.

The MW Measure HD board is installed in test head slot 2. A MW Synthesizer HD board must be internally connected to the MW Measure HD board, but the test head slot does not affect the channel map.

This workbook example contains the following components:

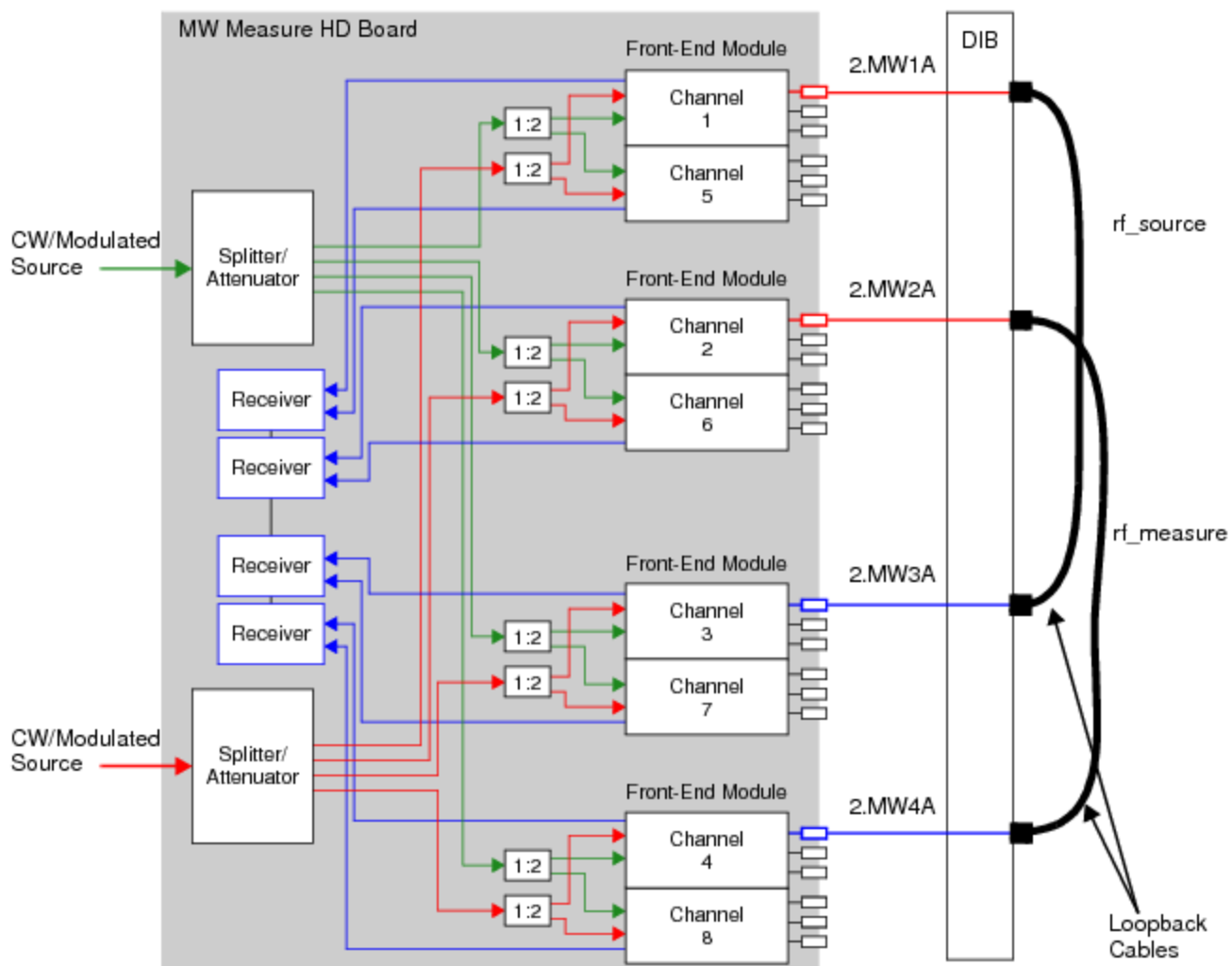
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|---|-----------------|
| • | DUT Connections |
| • | DataTool Sheets |
| • | VBCode |
| • | DSP Code |

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UltraWave24 Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : DUT Connections

DUT Connections



UltraWave24 MWSrc-MWReceiver Loopback Test Connections

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UltraWave24 Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : DataTool Sheets

DataTool Sheets

This workbook example includes the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Flow Table Sheet
- Test Instances Sheet
- Mixed Signal Timing Sheet
- Waveform Definition Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	rf_source	Analog
	rf_measure	Analog

Channel Map Sheet

Channel Map

DIB ID: View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
rf_source		MW	2.MW1A	2.MW2A
rf_measure		MW	2.MW3A	2.MW4A

Flow Table Sheet

Flow Table



Flow Domain:										
		Gate			Command					
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim
					Test	loopback_am_mod				
					Use-Limit	loopback_am_mod	carrier_pow	100		-20
					Use-Limit	loopback_am_mod	USB_pow	101		-20
					Use-Limit	loopback_am_mod	LSB_pow	102		-20

Test Instances Sheet

Test Instances

Test Procedure		
Test Name	Type	Name
loopback_am_mod	VBT	modsource_am

The test instance named loopback_am_mod has the following parameters and values as shown in the Instance Editor:

Name	Value	Comment
Sourcepin	rf_source	...
Capturepin	rf_measure	...
Sourcepower	-20	...
Sourcefreq	100000000	...
MWRecv_IF_Freq	5000000	...
AM_carrier_IF_Freq	2000000	...
Baseband_Freq	10000	...
modsource_FS	20000000	...
AM_waveform_size	20000	...
AM_Mod_Index	0.5	...
MWRecv_sample_rate	25000000	...
MWRecv_sample_size	25000	...

Mixed Signal Timing Sheet

The MW instrument does not use the Mixed Signal Timing sheet.

Waveform Definition Sheet

The MW instrument does not require a Waveform Definition sheet.

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UltraWave24 Examples : [MWModulatedSource-MWReceiver AM Signal Test](#) : VBT Code

VBT Code

```
Public Function modsource_am(ByVal Sourcepin As pinlist, _  
    ByVal Capturepin As pinlist, _  
    ByVal Sourcepower As Double, _  
    ByVal Sourcefreq As Double, _  
    ByVal MWRecv_IF_Freq As Double, _  
    ByVal AM_carrier_IF_Freq As Double, _  
    ByVal Baseband_Freq As Double, _  
    ByVal modsource_FS As Double, _  
    ByVal AM_waveform_size As Double, _  
    ByVal AM_Mod_Index As Double, _  
    ByVal MWRecv_sample_rate As Double, _  
    ByVal MWRecv_sample_size As Double)
```

' Parameter declarations

```
Dim CrestFactor As Double  
Dim baseband_signal As New DSPWave  
Dim LO_Signal As New DSPWave  
Dim modulation_signal As New DSPWave  
Dim Site As Variant  
Dim AM_capture_wave As New DSPWave  
Dim carrier_power As New SiteDouble  
Dim USB_power As New SiteDouble  
Dim LSB_power As New SiteDouble
```

Dim mwrecv_spectrum_reversed As New SiteBoolean

```
modulation_signal = AM_modulated_IF(AM_carrier_IF_Freq, Baseband_Freq, _
    AM_Mod_Index, modsource_FS, AM_waveform_size, CrestFactor)
```

' Add the Amplitude Modulated signal in DSPwave to the Wave Definitions

' Collection

If Not (TheExec.WaveDefinitions.Exists("AM_Wavedef")) Then

Call TheExec.WaveDefinitions _

.CreateWaveDefinition("AM_Wavedef", modulation_signal, False)

End If

' Set up MWModulatedSource

TheHdw.MWModulatedSource(Sourcepin).Signals.Add "am_signal"

TheHdw.MWModulatedSource(Sourcepin).Signals.DefaultSignal = "am_signal"

With TheHdw.MWModulatedSource(Sourcepin).Signals("am_signal")

.WaveDefinitionName = "AM_Wavedef"

.Type = tlMWSignalTypeIF

.IFFrequency = AM_carrier_IF_Freq ' this frequency is upconverted to the
center frequency

.ReverseSpectrum = True ' AM signal is symmetric, so this does not matter.

.SampleRate = modsource_FS

.SampleSize = AM_waveform_size

.CrestFactor = CrestFactor

.Frequency = Sourcefreq

.amplitude = Sourcepower

.LoadSettingsConnect

.Start tlStartImmediately

End With

' Set up the MWReceiver

With TheHdw.MWReceiver.Pins(Capturepin)

.Signals.Add "am_mod_cap"

.AmplitudeExpected = Sourcepower

.CenterFrequency = Sourcefreq

.IFFrequency = MWRecv_IF_Freq

```

.IFFilter = 10000000#
.Capture.SampleSize.Value = MWRecv_sample_size
.Capture.SampleRate = MWRecv_sample_rate

AM_capture_wave = .Signals("am_mod_cap").DSPWave.Pins(Capturepin)
mwrecv_spectrum_reversed = .IsSpectrumReversed(Sourcefreq)

.Connect
End With

' Trigger the MWReceiver
TheHdw.MWReceiver.Pins(Capturepin).Connect
TheHdw.MWReceiver.Pins(Capturepin).Signals("am_mod_cap").Trigger
TheHdw.MWReceiver.Pins(Capturepin).WaitForCaptureDone

' Retrieve the DSP wave handle, MWReceiverDSPWave

'-----
' For offline simulation only
' Import the saved DSP wave.
If TheExec.TesterMode = testModeOffline Then
    Dim ampl As Double
    ampl = 10# ^ ((Sourcepower + (CrestFactor - 3#) - 10#) / 20#)
    For Each Site In TheExec.Sites
        ' Should simulate spectrum reversal; AM is symmetric, however
        AM_capture_wave = AM_modulated_IF(MWRecv_IF_Freq, Baseband_Freq, _
            AM_Mod_Index, MWRecv_sample_rate, MWRecv_sample_size) _
            .Multiply(ampl)
    Next Site
End If
'-----

' AM_capture_wave(0).Plot "am_capture 0"
' AM_capture_wave(1).plot "am_capture 1"

' Call the DSP routine
Call rundsp.am_sb_power(AM_capture_wave, mwrecv_spectrum_reversed, _
    MWRecv_IF_Freq, Baseband_Freq, carrier_power, USB_power, LSB_power)

```

```

' Datalog the results
Call TheExec.Flow.TestLimit(resultval:=carrier_power, unit:=unitCustom, _
    customUnit:="dBm", ScaleType:=scaleNoScaling, _
    forcerevents:=tlForceFlow)
Call TheExec.Flow.TestLimit(resultval:=USB_power, unit:=unitCustom, _
    customUnit:="dBm", ScaleType:=scaleNoScaling, _
    forcerevents:=tlForceFlow)
Call TheExec.Flow.TestLimit(resultval:=LSB_power, unit:=unitCustom, _
    customUnit:="dBm", ScaleType:=scaleNoScaling, _
    forcerevents:=tlForceFlow)

' Disconnect the instruments
With TheHdw.MWModulatedSource.Pins(Sourcepin)
    .Signals.Stop tlStopImmediately
    .Disconnect
End With

TheHdw.MWReceiver.Pins(Capturepin).Disconnect

End Function

' The main parameters passed into this function are:
'
' IFFreq - the Amplitude Modulated signal Carrier IF Frequency.
' modulation_Freq - frequency at which the amplitude is modulated
' (i.e., the AM signal frequency)
' sample_rate - the sample frequency (in Hz) of the modulated segment
' sample_size - the sample size of the modulated segment
' modulation_index - the Modulation Index for the Amplitude Modulated signal
'
' The peak amplitude is normalized to remain in the +/-1.0 range
'
' The function returns
'
' crest_factor_dB: ratio of peak to RMS power, in dB
'

```

```

Public Function AM_modulated_IF(ByVal IFFreq As Double, _
                                ByVal modulation_freq As Double, _
                                ByVal modulation_index As Double, _
                                ByVal sample_rate As Double, _
                                ByVal sample_length As Long, _
                                Optional ByRef crest_factor_dB As Double) _
    As DSPWave

    Dim baseband_signal As New DSPWave
    Dim carrier_signal As New DSPWave
    Dim modulation_signal As New DSPWave

    ' Create AM baseband "information" signal: will modulate the carrier
    '   amplitude. Determines the sideband spacing of the modulated waveform.

    Call baseband_signal _
        .CreateCos(2 * PI * modulation_freq / sample_rate, 0, sample_length)
    baseband_signal.SampleRate = sample_rate

    ' Create carrier signal: sets the center frequency of the AM waveform

    Call carrier_signal.CreateCos(2 * PI * IFFreq / sample_rate, 0, sample_length)
    carrier_signal.SampleRate = sample_rate

    ' Create the AM signal by multiplying the two sine waves together
    ' A[1+(mod_index)cos(baseband_freq)]cos(carrier_freq) or
    ' A[cos(carrier_freq)+(cos(carrier_freq)*cos(baseband_freq)*mod_index]

    modulation_signal = baseband_signal.Multiply(modulation_index)
    modulation_signal = _
        carrier_signal.Multiply(modulation_signal).Add(carrier_signal)

    ' Normalize to +/-1.0
    Dim peak_ampl As Double
    peak_ampl = modulation_signal.CalcMaximumMagnitude
    modulation_signal = modulation_signal.Multiply(1# / peak_ampl)

    modulation_signal.SampleRate = sample_rate

```

```
' Calculate the CrestFactor of the AM signal
crest_factor_dB = 20 * Log(modulation_signal.CalcMaxMagnitude / _
    modulation_signal.CalcRMS) / Log(10#)
Set AM_modulated_IF = modulation_signal
```

End Function

DSP Code

```
Public Function am_sb_power(ByVal Capture As DSPWave, _
    ByVal spectrum_reversed As Boolean, _
    ByVal IFFreq As Double, _
    ByVal Freq_BB As Double, _
    ByRef carrier_dBm As Double, _
    ByRef power_dBm_hi As Double, _
    ByRef power_dBm_lo As Double) As Long
```

```
' Parameter Declarations
```

```
Dim bin_hi As Long, bin_lo As Long, bin_carrier As Long
Dim Fres As Double, ampl_hi As Double, ampl_lo As Double
Dim ampl_carrier As Double
Dim bin_spacing As Long
```

```
' Calculate Frequency resolution of the capture
```

```
Fres = Capture.FrequencyResolution
bin_carrier = IFFreq / Fres
bin_spacing = Freq_BB / Fres
```

```
' Calculate the bin separation of the base band signal
```

```
' Calculate the bin number of the upper and lower side band
```

```
If spectrum_reversed Then
    bin_hi = bin_carrier - bin_spacing
    bin_lo = bin_carrier + bin_spacing
Else
    bin_hi = bin_carrier + bin_spacing
    bin_lo = bin_carrier - bin_spacing
End If
```

' Calculate the Amplitudes of the LSB and USB.

```
ampl_carrier = Capture.CalcAmplitudeFromTimeDomain(bin_carrier)
ampl_hi = Capture.CalcAmplitudeFromTimeDomain(bin_hi)
ampl_lo = Capture.CalcAmplitudeFromTimeDomain(bin_lo)
```

' Convert the Amplitudes calculated above to power.

```
carrier_dBm = (10# / Log(10#)) * _
               Log(1000# * ampl_carrier * ampl_carrier / (2 * 50))
power_dBm_hi = (10# / Log(10#)) * Log(1000# * ampl_hi * ampl_hi / (2 * 50))
power_dBm_lo = (10# / Log(10#)) * Log(1000# * ampl_lo * ampl_lo / (2 * 50))
```

am_sb_power = 0 ' TL_SUCCESS

End Function

UW24Subsys_Cal.3.1

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UltraWave24 Instrument Calibration

UltraWave24 Instrument Calibration

The MW instruments require calibration to meet performance specifications. Instrument specifications depend on all three calibration levels supported by the UltraFLEX test system:

Level 1	Internal calibration (also called automatic calibration or autocal)
Level 2	Maintenance programs
Level 3	Internal reference calibration to traceable external standards

Most MW Level 3 calibrations are performed at the factory before shipment. Level 1 and 2 calibrations are performed in the field, in the environment where the tester is used. In addition to the three levels of calibration, the MWSource and MWReceiver instruments perform calibrations at run time. Run-time calibrations include autocal and program-specific levelling. The three calibration levels and the UltraFLEX calibration strategy are described in [Levels of Calibration](#) and in [Maintenance Environment Help](#). This section includes the following topic:

- [Run-Time Calibration](#)

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UW24Subsys_Cal.3.2

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UltraWave24 Instrument Calibration : Run-Time Calibration

Run-Time Calibration

The MWSource and MWReceiver meet specifications by automatically performing calibrations at run time. These run-time calibrations are:

- Source Level Calibration (Leveling)
- Receiver Gain Calibration

Source Level Calibration (Leveling)

Source level calibration, also called “leveling,” finely calibrates the MW sources (MWSource, MWModulatedSource, and MWMultiToneSource) at each frequency set in a program. Leveling takes place at run time.

Leveling recalibrates the reference power level for a frequency and channel using the internal port power sensor and the factory calibration data. The source then uses the reference cal data and other calibration data to determine source settings for all other levels at that frequency.

Leveling occurs the first time a specific frequency is set for the MWSource or any other MW source type. Later VBT statements setting the same frequency use cached data and do not level, unless leveling is invoked using the Pins.LevelImmediate VBT property for the source type. For example, see the LevelImmediate property for the Pins interface of TheHdw.MWSource. Leveling can also be invoked for all frequencies used in a test program using the Pins.LevelRefresh VBT method for a source.

Receiver Gain Calibration

Receiver gain calibration, or system gain calibration, finely calibrates the MWReceiver gain for a specific instrument setup at one input frequency, at run time.

Receiver gain calibration connects the receiver to the source through an internal loopback path and transfers an absolute reference level determined during factory cal to the receiver. The gain for the path is calibrated to the receiver input. Receiver gain is measured for each combination of parameter settings for full instrument accuracy.

Only the multiplexer that includes the channel and port for the assigned pin is calibrated.

As with source leveling, receiver gain calibration occurs the first time a specific instrument setup is used for the MWReceiver. Later VBT statements with the same setup use cached data and do not run calibration. Gain calibration can also be invoked for all setups used in a test program using the Pins.LevelRefresh VBT method for the receiver.

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UW24Subsys_Displays.5.1

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UltraWave24 Displays

UltraWave24 Displays

The UltraWave24 subsystem supports several displays for test programming and debugging:

Microwave Board Debug Display	This display provides physical board-level views of the Microwave instrument hardware (boards) installed and configured in the tester. It shows the boards, how they are connected, and how the hardware functions are used by logical instruments. The Microwave Board Debug Display is available in TDE.
Microwave Debug Display	This display shows the state of one or more MW logical instruments in the context of a site. Each debug display features the control and display functions of a logical instrument, but hides the physical hardware. The Microwave Debug Display is available in TDE.
Smith Display	The Smith Display plots s-parameter data on a Smith Chart, mapping coefficients of reflection to impedance. You create s-parameter data with the MWVNA logical instrument. It is available from the Windows command line. See Smith Display.

This section describes features specific to the UltraWave24 Microwave displays. For more information on using debug displays, see the following:

- UltraWave24 Microwave Board Debug Display
- UltraWave24 Microwave Debug Display

- [Using Debug Displays in TDE](#)
-
- [Producing Code from Debug Displays](#)

IG-XL also includes the Vector Scope, a display tool that plots complex data. This display is useful for analyzing modulated RF signal waveforms. Vector Scope is available from either the Windows command line or by using the DSPWave Plot method in IG-XL debug mode. For more information, see [Vector Scope](#).

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UW24Subsys_Displays.5.2



UltraWave24 Displays : UltraWave24 Microwave Board Debug Display

UltraWave24 Microwave Board Debug Display

The Microwave Board Debug Display for the UltraWave24 subsystem provides two kinds of board level views of the instrument hardware in a tester. The different views are displayed, one at a time, in the display window as tabs, one tab for each view:

Navigator	This tab shows the Synthesizer and Measure boards in the tester configuration as blocks with internal tester connection terminals. You can read and display connectivity information, launch board views, and navigate to the different board views. The display can only include one Navigator tab.
Board view	This tab shows a high-level diagram of a board with a high-level schematic diagram of the main board functions with signal routing and settings. The display can include more than one board view, each with a tab. You can read and display the primary settings, basic signal paths, and connections for logical instruments. A board view has links to the logical instrument debug displays. The Measure board tab is useful for examining interactions between logical instrument paths. For example, you can view and compare the signal paths for a MWModulatedSource and a MWReceiver.

The board display shows only the hardware in your tester configuration. Displays are updated to reflect the current logical instrument connections and settings. In some cases, you can change instrument settings from board displays.

For more information, see the following topics:

- [Starting the UltraWave24 Microwave Board Debug Display](#)
- [Navigator Tab and Controls](#)
- [Board Tabs and Controls](#)

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UltraWave24 Displays : [UltraWave24 Microwave Board Debug Display](#) : Starting the UltraWave24 Microwave Board Debug Display

Starting the UltraWave24 Microwave Board Debug Display
Before you can start the [Microwave Board Debug Display](#), the active program must:

- | | |
|---|--|
| • | Be in debug mode |
| • | Use at least one Microwave pin |
| • | Be validated |

To launch the [Microwave Board Debug Display](#), on the [IG-XL Tools](#) tab, in the Launch group, select Subsystems>Microwave Board. The Microwave Board Debug Display appears with the Navigator tab shown. See [Navigator Tab and Controls](#).

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UW24Subsys_Displays.5.4

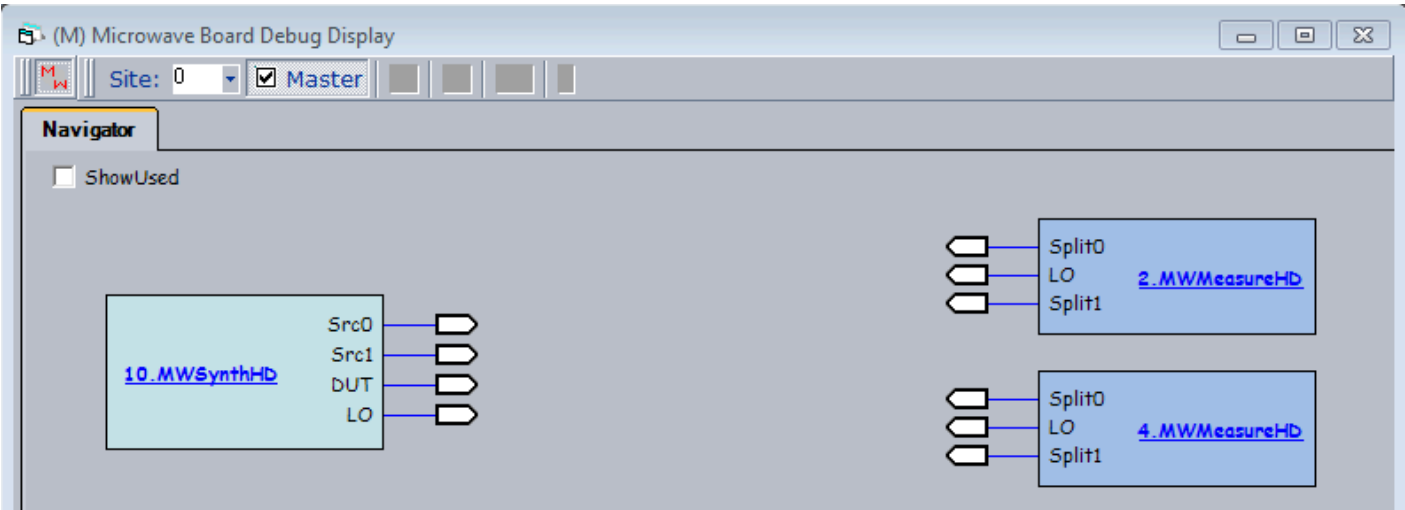
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UltraWave24 Displays : UltraWave24 Microwave Board Debug Display : Navigator Tab and Controls

Navigator Tab and Controls

When it opens, the Microwave Board Debug Display shows the Navigator tab. The example in the figure shows a configuration with one MWSynthesizerHD board and two MWMeasureHD boards.



Board Debug Display Navigator Tab

The display shows all boards in the current tester configuration arranged in two columns as follows:

- Synthesizer boards on the left side
- Measure boards on the right side
- Both board columns in ascending order top to bottom, by slot number

For each board, the Navigator tab shows:

- Slot number

- Board type
- Internal port names and terminals

The Navigator tab has the following controls:

board block	Represents a board. The tooltip shows a list of all connected boards.
board block label	Identifies the slot number and UltraWave24 board type. Also, as an active link (underlined), the board label does one of the following: - Adds a new board view with tab, if there is not a board tab for the board in that slot. - Selects the tab for the board, if there is already a board tab for the board in that slot.
terminal symbol	Shows internal cable connections between Synthesizer boards and Measure boards. Terminals are active. Click to show connections between a Synthesizer board output and all the inputs of connected Measure boards. The display clears any existing connections before showing new ones. The DUT output of a Reference Synthesizer (MWSynthRef) provides external signals to the DUT; external connections are not shown.
Show Used	Displays only boards used in the current channel map, when checked. Otherwise, the display shows all boards in the current tester configuration.

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UW24Subsys_Displays.5.5



UltraWave24 Displays : [UltraWave24 Microwave Board Debug Display](#) : Board Tabs and Controls

Board Tabs and Controls

When the Microwave Board Debug Display first opens, it shows only the Navigator tab.

To create a new tab with a board view from the Navigator tab:

- Click the label link (underlined) on the block for the board. For example, [2.MWMeasureHD](#) is the label for a Measure board in slot 2.

A new tab appears with the board label, such as [2-MWMeasureHD](#), and the high-level schematic for the board type appears.

If a tab for the board already exists, when you click the label link in the block, the display changes to the tab for the labeled board.

To view an existing board tab from the Navigator tab, do either of the following:

- Click the label link (underlined) on the block for the board. For example, [2.MWMeasureHD](#) for a Measure board in slot 2.

The display changes to the tab for the labeled board.

- Click the tab for the board.

The display changes to the tab for the board.

To close a board tab:

- Click the red "X" button at the far right on the tab bar, while the tab is selected.

The tab disappears. You can close only board tabs, not the Navigator tab.

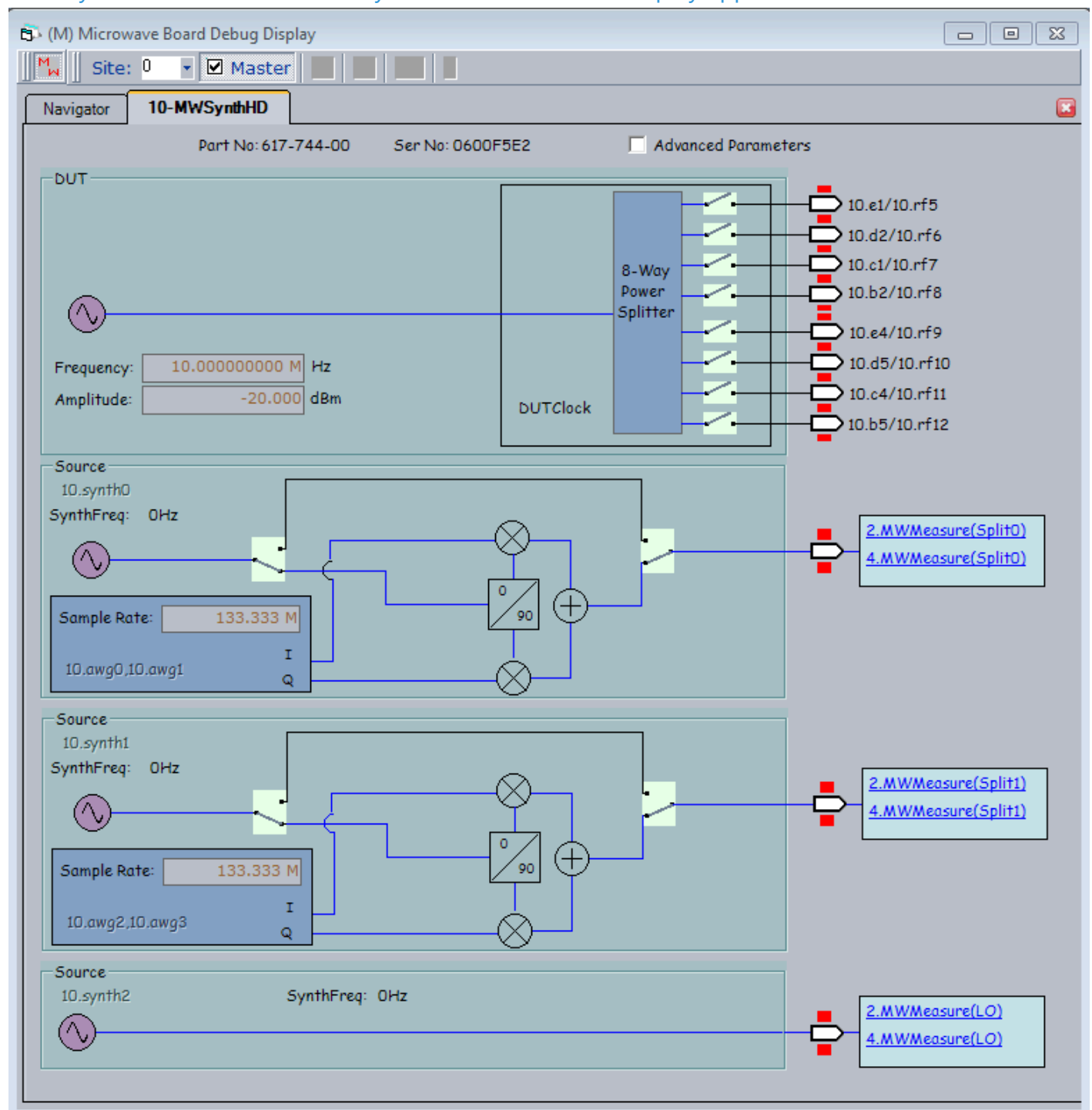
The information displayed on a board tab and the controls are specific to the board type:

- Synthesizer HD Board Tab Display and Controls

- Measure HD Board Tab Display and Controls

Synthesizer HD Board Tab Display and Controls

When you create or select a MWSynthHD board tab, the display appears:



MWSynthHD Board Tab

The display shows synthesizers and sources on a MW Synthesizer HD board, along with the primary settings, basic signal paths, and connections.

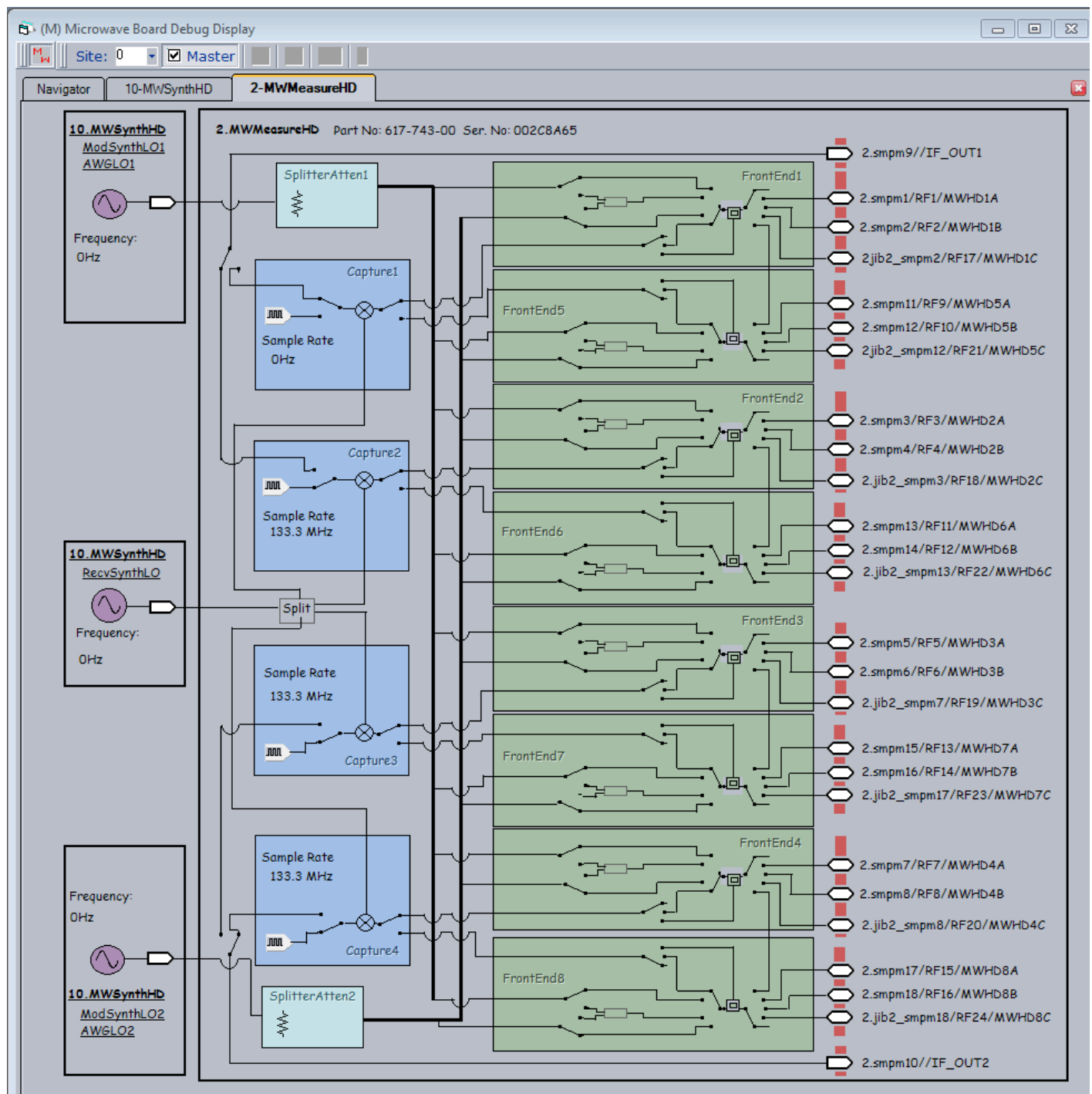
The DUT Clock shows output terminals at the DIB interface (RF ports or pins). The CW/modulated sources show the internal output connection to a splitter input on one or more Measure HD boards. The LO source shows the internal output connection to a splitter input on one or more Measure HD boards. Up to four Measure HD boards can be connected and shown.

The Synthesizer HD Board Tab has the following controls:

Frequency	Sets and displays the sourced signal frequency for the DUT Clock signals at one to eight DIB pins. The tooltip for this field shows the allowable frequency range.
Amplitude	Sets and displays the sourced signal amplitude for the DUT Clock signals at one to eight DIB pins. The tooltip for this field shows the allowable amplitude range.
DUT Clock output terminal	Tooltip shows pin name and logical instrument name, only if instrument is connected.
SynthFreq	Displays the synthesizer frequency for the CW/modulated and LO sources.
source output terminal	Displays names of connected Measure boards. Also, as an active link (underlined), the board label selects the tab for the measure board.
Sample Rate	Displays the sample rate of the internal I and Q modulation signals.
relays	Displays the connection state for: • Individual DUT Clock outputs connected or disconnected • Source set to either CW or modulated
Advanced Parameters	Displays detailed information about the internal states of some MW hardware settings, when checked. This information is only useful during support from Teradyne.

Measure HD Board Tab Display and Controls

When you create or select a MWMeasureHD board tab, the display appears:

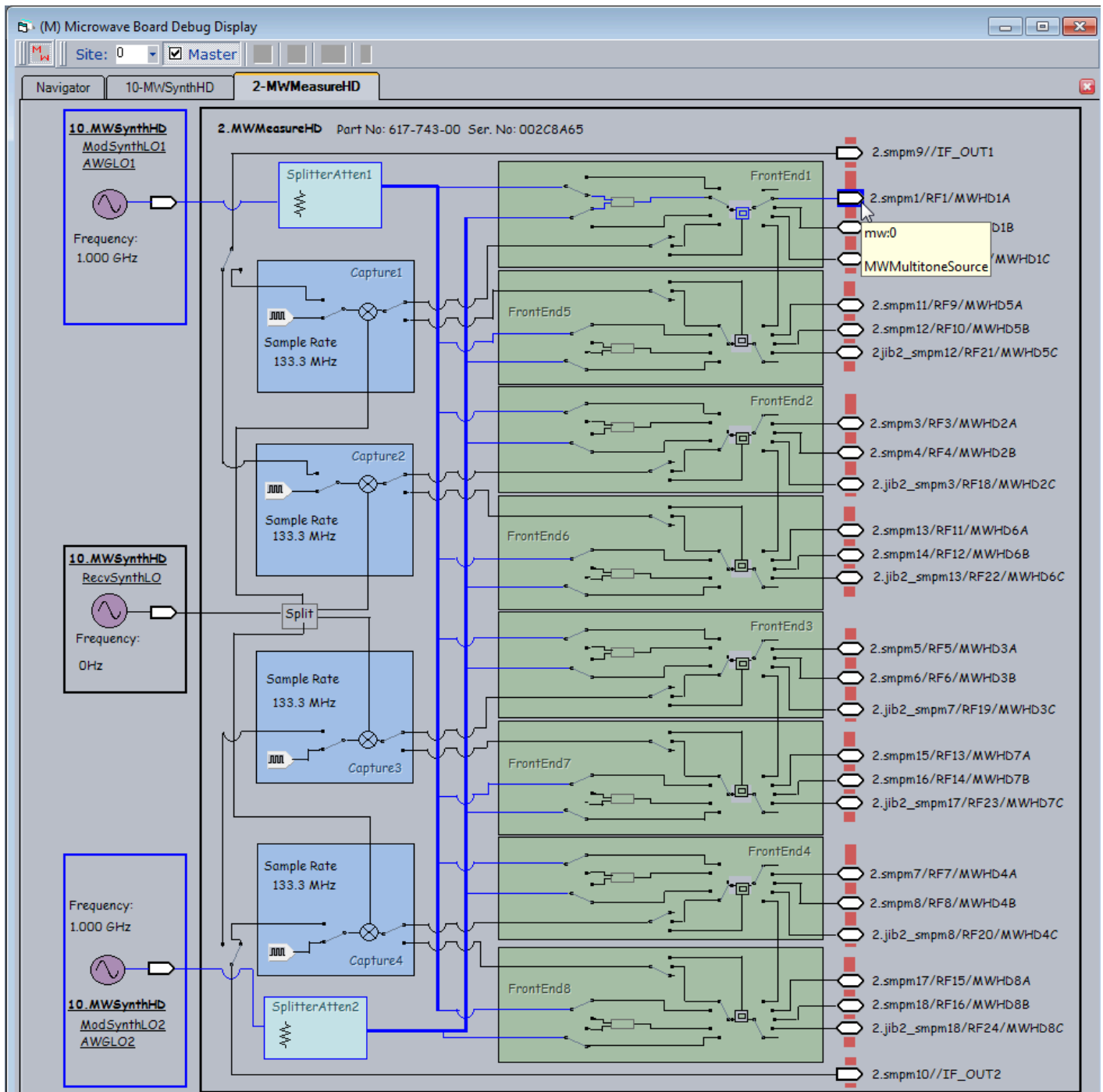


MWMeasureHD Board Tab, No Logical Instruments Connected

The display shows the four front ends (MW channels), four receivers with digitizers, and the two splitter/attenuators for a MW Measure HD board. In addition, the display shows the synthesizers providing source and LO signals to the Measure board. As with the Synthesizer boards, the Measure board display shows the primary settings, basic signal paths, and connections.

The MW Measure HD board tab shows bidirectional terminals representing the multiplexer RF ports at the DIB interface, six ports for each front end. The tab also shows the IF output port for each capture block (receiver/digitizer).

When instruments are connected, the display shows more information as shown in the next figure.



Measure Board Tab, Logical Instrument Connected to DUT

In this example, the MWMultiToneSource is connected. In this case, with the site set to the master site (site 0), the Measure board tab shows the following primary settings, signal paths, and DIB connections for connected ports:

- Signal paths through the front-end blocks and into the capture blocks using switch positions, colored signal wires.
- Connected RF port using highlighting: a blue background block behind the terminal symbol or symbols for the selected site, and a black background block behind the terminal symbol or

- symbols for any other sites.
- Connected RF port signal direction by changing from bidirectional input/output symbol (points both right and left) to. I this example, the output symbol (points right) is shown for source direction for the MWMultiToneSource
- Connected pin name and logical instrument name for each connected RF port terminal though a tooltip (hover pointer over symbol).

A Measure Board tab has the following controls:

Frequency	Sets and displays the frequency for a source instrument (MWSource, MWMultiToneSource, and MWModulatedSource) and MWReceiver instrument when an assigned pin is connected. The tooltip for this field shows the allowable frequency range. In a synthesizer board block, displays the frequency setting of the synthesizer in the block.
Amplitude	Sets and displays the sourced signal frequency for the DUT Clock signal at one to four DIB pins. The tooltip for this field shows the allowable amplitude range.
DIB port input or output terminal	Indicates the connection state and port direction for a connected pin, and selects the logical instrument Debug Display for the selected site. For a connected pin: - Terminal symbol is highlighted (blue background block for the selected site or black background block for any other site). - Tooltip shows pin name and logical instrument name. - Click a terminal for the selected site (blue background) to start the Microwave Debug Display and display the logical instrument for the connected pin.
board block label	Identifies a synthesizer slot number and synthesizer board type. Also, as an active link (underlined), the synthesizer board label does either of the

following:

- Adds a new board view with tab, if there is not a board tab for the board in that slot.
- Selects the tab for the board, if there is already a board tab for the board in that slot.

		
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UltraWave24 Displays : UltraWave24 Microwave Debug Display

UltraWave24 Microwave Debug Display

The Microwave Debug Display is an interactive tool that shows and controls the state of one or more MW logical instruments in the context of a site. This multi-instrument display overlays the physical hardware of the MW subsystem, featuring the programmable and read-only parameters of the logical instruments.

You can add or remove instruments from the display, as well as expand or collapse instruments on the display to save screen space.

The Microwave Board Debug Display can show instrument settings and resource constraints when you use it with the Microwave Debug Display.

The following topics describe the display controls and how to use the display:

- | | |
|---|--|
| • | Starting the UltraWave24 Microwave Debug Display |
| • | UltraWave24 Microwave Debug Display Controls |
| • | UltraWave24 Logical Instrument Display Controls |

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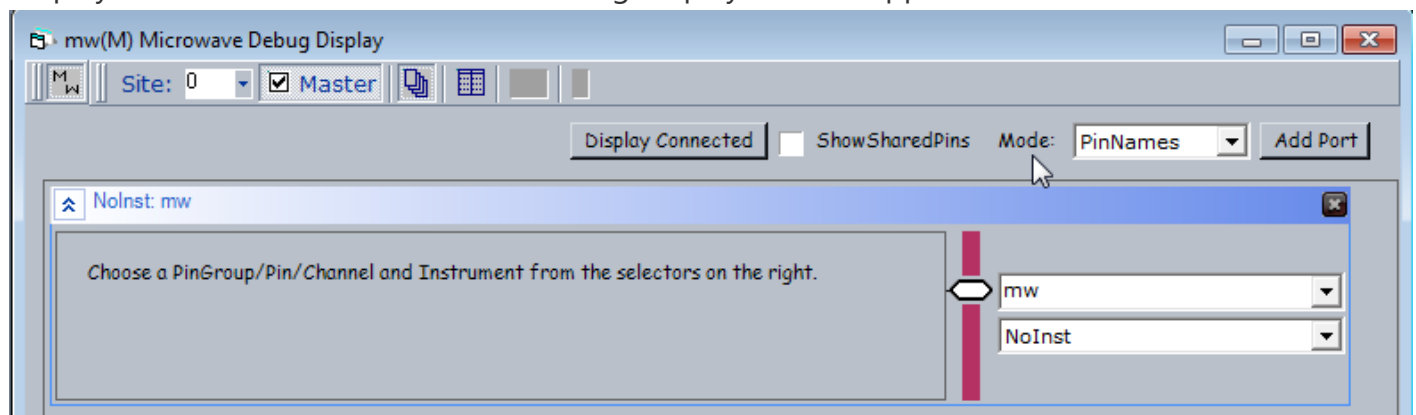
UltraWave24 Displays : UltraWave24 Microwave Debug Display : Starting the UltraWave24 Microwave Debug Display

Starting the UltraWave24 Microwave Debug Display

Before you can start the Microwave Debug Display, the active program must:

- Be in debug mode
- Use at least one Microwave pin, type MW or MWSource
- Be validated

To launch the Microwave Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>Microwave. The Microwave Debug Display window appears.



Microwave Debug Display Startup, No Instruments Connected

At startup, the Microwave debug display shows the instruments currently connected by the test program. If no instruments are connected, the display appears showing an empty instrument port (block and terminal symbol) and prompts you to choose a pin and instrument from the (unlabeled) pin and instrument type drop-down lists just to the right side of the port symbol.

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UltraWave24 Displays : UltraWave24 Microwave Debug Display : UltraWave24 Microwave Debug Display Controls

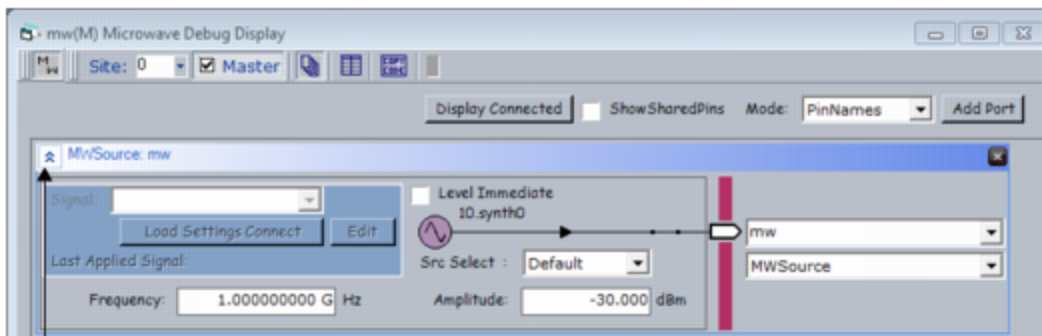
UltraWave24 Microwave Debug Display Controls
The Microwave Debug Display contains three categories of controls:

- TDE Controls
- General Controls
- Logical Instrument Controls

TDE Controls
The TDE Site selector control and the Apply All Sites, Summary Display, and Copy Generated Code buttons all appear on the Microwave Debug Display toolbar. These controls have the same functions for all debug displays. See Using Debug Displays in TDE.
General Controls
Use the general Microwave Debug Display controls to:

- Add logical instrument ports to the debug display.
- Remove logical instrument ports from the debug display (except for the last port: one must be displayed).
- Expand or collapse an instrument port.

- Set the display mode to pin names or channel names.
- Assign DUT pins or MW channels to a specific instrument type.
- Display all connected instruments in a standard order.
- Show shared pins or shared channels in the instrument port for all connected instruments.
- Identify and locate shared resource conflicts.



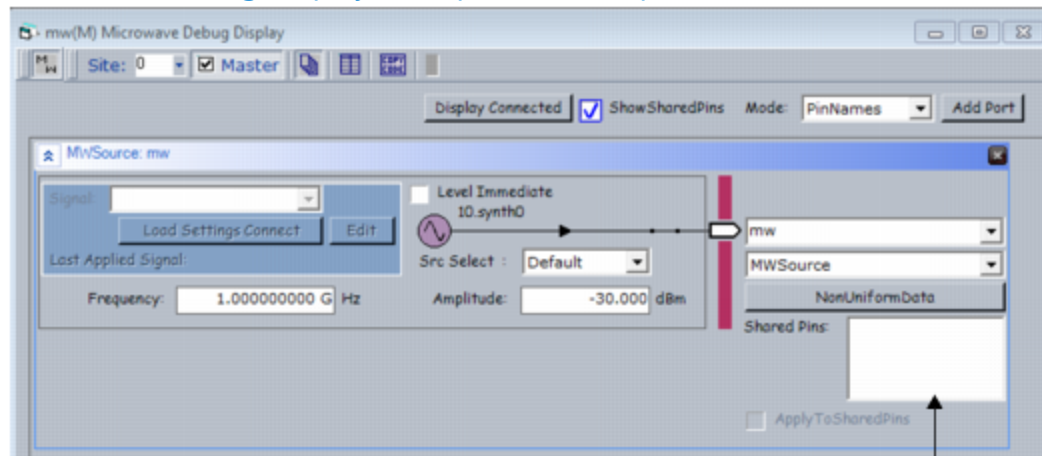
Collapse Icon

Microwave Debug Display, Expanded, with Collapse Icon



Expand Icon

Microwave Debug Display, Collapsed, with Expand Icon



Shared Pins Field

Microwave Debug Display, Expanded, with Shared Pins Field

The source in the example windows above does not use a Signal to control amplitude and frequency.

Logical instrument controls appear on a logical instrument port, which is connected through a switch to a MW port. You use logical instrument controls to read and write logical instrument parameters and connections.

The following general controls are on the UltraWave24 Microwave Debug Display.

title bar	Displays the selected logical instrument name and pin name.
Display Connected	Displays and sorts the instrument ports, as follows: • Sorts and displays the ports for connected instruments. Ports are shown expanded at the top of the display with sources first, followed by receivers, and then the DUT references. Ports with the same instrument type are sorted by pin groups, then pins, then channels. • Adds a port for each instrument connected in VBT code, if it doesn't have an existing port display. These ports are sorted as listed as above. • Displays ports for unconnected instruments at the bottom of the display. Unconnected ports are collapsed and unsorted.
ShowSharedPins ShowSharedChans	Displays the labeled shared pins field or shared channels field for each instrument port, when checked. Also displays the NonUniformData button and the ApplyToSharedPins or ApplyToSharedChannels check box. (The check box is disabled when there are no shared pins.) The field displays pins for connected ports that share hardware resources.
NonUniformData	Starts an non Uniform Data Summary window and displays nonuniform data for a connected port, only if data is available.

ApplyToSharedPins	Applies changed settings for a port to all pins or channels that share the same hardware resources.
ApplyToSharedChannels	Only displayed when ShowSharedPins or ShowSharedChans is checked.
Mode	Sets the type of data listed on the Pins drop-down list for each logical instrument port displayed. Select from the drop-down list: • PinNames: Lists pin names for all channels mapped to a pin and channel names for all channels mapped to a pin. Pin group names are listed last, in bold. • Channels: Lists channel names for all channels.
Add Port	Adds a logical instrument block and port to the display. The port shows the first pin on the pin name drop-down list. The instrument block remains generic until you select an instrument type from the instrument type drop-down list. A scrollbar appears if there are more ports than the window can display. The maximum number of ports is 128.
pin or channel name	Sets and displays the DUT pin name, pin group name, or the MW channel name associated with the instrument port. Names on the drop-down list depend on the Mode control setting. If Mode is set to PinNames, the list displays pin group names at the end of the list, in bold.
instrument type	Sets and displays the logical instrument type assigned to a port. Select from the drop-down list: • NoInst: No logical instrument is selected. • MWModulatedSource • MWMultiToneSource • MWSource

- [MWNoiseSource](#)
- [MWReceiver](#)
- [MWVNA](#)

When you select a logical instrument, the debug display updates the controls to reflect the state of the instrument.

When a pin is defined in the channel map with tester channel type [MWSource](#) for MW Reference DUT resources and selected in the pin list, only [MWSource](#) appears in the instrument list.

See [UltraWave24 Logical Instrument Display Controls](#).

DIB terminal symbol

Indicates the port direction for the selected instrument type. The port direction can be input (points to left), output (points to right), or bidirectional (points to both left and right).

The port direction can change depending on the instrument type selected.

expand/collapse

Expands or collapses the logical instrument display. Click the title bar to collapse or expand the logical instrument port display. The icon changes to indicate the click action.

Expanding or collapsing an instrument block does not change the instrument connections.

remove port button (small x on title bar)

Removes a logical instrument block and port from the display.

Since at least one instrument block must remain on the Microwave Debug Display at all times, the remove button is inactive when only one block is displayed.

This button does not change the instrument connections.

resource conflict

Displays a bar-through-circle ("not allowed") symbol over a connect relay when you connect an instrument. The symbol indicates that you cannot connect the instrument because it will conflict with another instrument.

The relay is disabled. You must resolve the conflict by disconnecting the instrument with the blocking shared resource before you can connect the instrument.

If you point at the relay, a Resource Conflict balloon opens and stays open until you close it.

For the active site, the balloon displays a link to the conflicting pin or channel. The link navigates to the conflicting instrument port. The display adds a port for the conflicting instrument if one is not already open.

Displays resource conflicts with for other sites if `ApplyAllSites` is set, but without a link to the conflicting pin.

Logical Instrument Controls

Each instrument block displays specific information for the instrument type selected. For a list of logical instrument types, see [UltraWave24 Logical Instrument Display Controls](#).

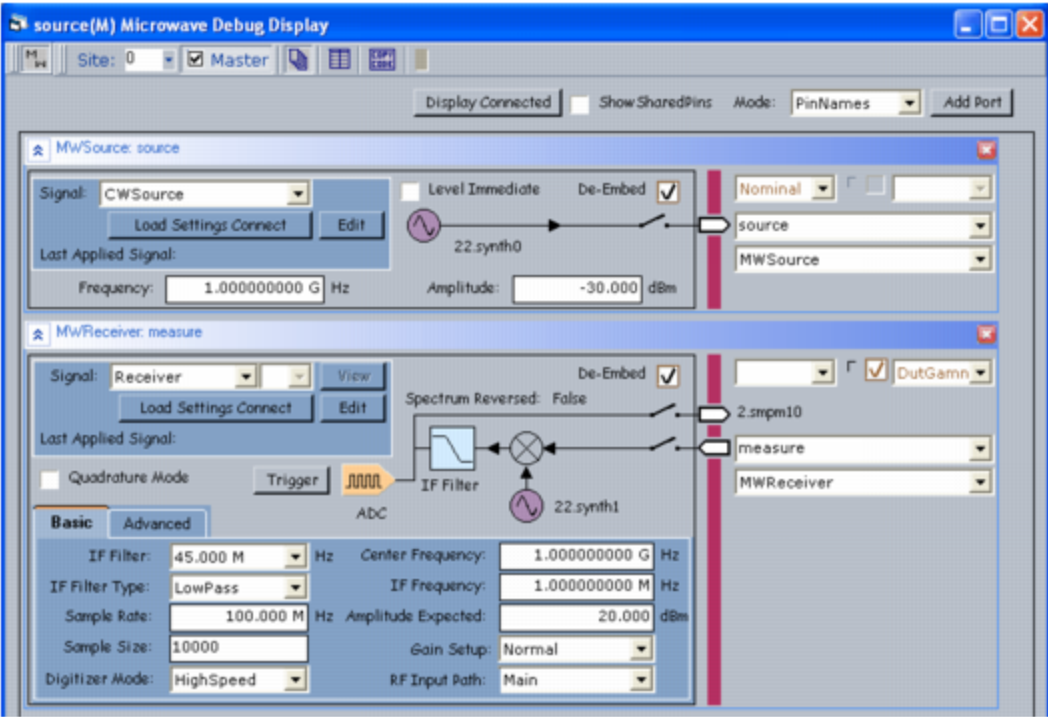
MWDIB Controls for Deembedding

Test programs can include test fixture (DIB) s-parameter data and DUT Gamma information for deembedding one-port and two-port s-parameter measurements. This information is contained in the MWDIB VBT object. If you are using MWDIB in your test program, additional controls in the displays appear. MWDIB controls can be in any of the following states:

- If the test program does not contain any MWDIB syntax, MWDIB controls are hidden in the displays.
- If the MWDIB is used in the test program, but the specified pin has no data, MWDIB information is grayed out in the displays.
- If the MWDIB is used in the test program and the specified pin has data, then MWDIB information is displayed and is interactive.

The following figure shows UltraWave24 Source and Receiver with MWDIB controls showing Deembed and DUT Gamma information.

For more information, see [MWDIB](#) in the VBT Syntax Reference.



Microwave Displays Showing Deembed Controls

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UltraWave24 Displays : [UltraWave24 Microwave Debug Display](#) : [UltraWave24 Logical Instrument Display Controls](#)

UltraWave24 Logical Instrument Display Controls

Each Microwave logical instrument block shows a high-level schematic representation of the instrument. It also provides controls which allow you to set the parameters and perform the operations provided by the Pins VBT language for that instrument.

Each block represents a single logical instrument that is connected to a pin or channel. For descriptions of port display controls for the MW24 logical instruments, see the following sections:

- [UltraWave24 MWModulatedSource Debug Display](#)
- [UltraWave24 MWMultiToneSource Debug Display](#)
- [UltraWave24 MWNoiseSource Debug Display](#)
- [UltraWave24 MWReceiver Debug Display](#)
- [UltraWave24 MWSource Debug Display](#)
- [UltraWave24 VNA Debug Displays](#)

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UltraWave24 Instrument Programming

UltraWave24 Instrument Programming

At the instrument code level, all MW instruments are programmed in the same general way. As with other kinds of UltraFLEX instruments, you use the VBT language and debug display for each individual MW logical instrument to program the functions of the MW subsystem. In addition, you must connect MW tests to the structure of an IG-XL test program.

When you program several logical instruments, IG-XL checks for resource conflicts at run time. Most conflicts are due to hardware resource mapping and resource sharing. See [Architecture and Resource Mapping](#) and [Shared Hardware Functions](#).

UltraWave24 instruments do not interact by taking resources currently in use by another instrument. IG-XL run-time errors notify you of resource conflicts and require you to resolve them. This following topics are available:

- [Simulated Configuration Files](#)
- [IG-XL Programming for UltraWave24 Instruments](#)
- [Using UltraWave24 in Concurrent Testing](#)
- [UltraWave24 Pattern Control](#)

The following sections describe how to program each MW logical instrument:

- [UltraWave24 MWModulatedSource Programming](#)

- [UltraWave24 MWMultiToneSource Programming](#)
- [UltraWave24 MWNoiseSource Programming](#)
- [UltraWave24 MWReceiver Programming](#)
- [UltraWave24 MWSource Programming](#)
- [UltraWave24 VNA Programming](#)

For sample MW code, see [UltraWave24 Examples](#).

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UltraWave24 Instrument Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the UltraWave24 instruments offline by forcing IG-XL to map the UltraWave subsystem. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps UltraWave24 boards in slots 2 and 10. This configuration has fully loaded and MWMeasureHD and MWSynthHD boards.

<Version>

1

</Version>

<TEST_HEAD TH 1>

#slot[.subslot]	Type	idprom (type serial rev company)
2.0	MWMeasureHD	617-743-00 002C8C1A 1408-A 5445
	MWMeasureHD	618-489-01 03400745 1406-A 5445
	FrontEnd1	618-558-01 0D493017 1406-A 5445
	FrontEnd2	618-558-01 0D493006 1406-A 5445
	FrontEnd3	618-558-01 0D493005 1406-A 5445
	FrontEnd4	618-558-01 0D493003 1406-A 5445
	SplitterAtten	618-617-02 034007F3 1406-A 5445

	Receiver	617-419-02 034006F4 1406-A 5445
	Digitizer	615-570-01 0306BF1E 1349-A 5445
	PowerDistribution	615-568-01 03400757 1407-A 5445
	PowerBoard	617-560-01 0340078D 1345-A 5445
10.0	MWSynthHD	617-744-00 002C8C19 1408-A 5445
	MWSynthHD	618-489-01 034005B4 1406-A 5445
	DividerModulator1	618-560-01 0D000034 1405-A 5445
	DividerModulator2	618-560-01 0D00003A 1405-A 5445
	DividerModulator3	618-560-01 0D000043 1405-A 5445
	AWG	615-566-01 034007D9 1401-A 5445
	DutClock	616-673-01 0340075C 1405-A 5445
	BaseBand	615-573-02 0310104E 1405-A 5445
	PowerBoard	617-560-01 0340078A 1345-A 5445
24.0	SupportBoard	974-221-03 c0e11bb 1032-B 5445
	SupportBoard	956-416-30 c0e0b8d 0944-B 5445
62.0	DistributionBoard	956-455-00 300b04c 0508-A 5445
	DistributionBoard	956-455-00 300b04c 0508-A 5445

</TEST_HEAD TH 1>

The simulated configuration file also requires a MWSynth cable section. The following example shows an UltraWave 24 Synth board cabled to a Measure board.

<CABLE MWSYNTH 1>

#Sythesizer Channel	Measure Channel
#<THSlot>.<MWSynth Out>	<THSlot>.<MWMeasure In>
10.LO	2.LO
10.CW/ModSrc1	2.CW/ModSrc1
10.CW/ModSrc2	2.CW/ModSrc2
10.LO	4.LO
10.CW/ModSrc1	4.CW/ModSrc1
10.CW/ModSrc2	4.CW/ModSrc2

</CABLE MWSYNTH 1>

To use [UltraWave24 instruments offline](#), the `SimulatedConfig.txt` file must include every board needed in the IG-XL workbook. The `SimulatedConfig.txt` file represents the state of the offline tester. To use [UltraWave24 instruments online](#), you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see [Tester Configuration](#).

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UltraWave24 Instrument Programming : IG-XL Programming for UltraWave24 Instruments

IG-XL Programming for UltraWave24 Instruments

You create MW tests (test functions using MW logical instruments) using the VB editor to code VBT functions in VB modules. MW tests become part of a test program when you include them in a worksheet in an IG-XL workbook. MW tests require the following data, typically entered using DataTool:

- | | |
|--|--|
| | • DUT pin data in a Pin Map sheet. See Programming the Pin Map. |
| | • Tester channel data in a Channel Map sheet. See Programming the Channel Map. |
| | • Test instance data in a Test Instances sheet |
| | • Test names and execution flow control data in the Flow Table sheet |

For more information about IG-XL workbooks, worksheets, and data, see About DataTool.

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UltraWave24 Instrument Programming : IG-XL Programming for UltraWave24 Instruments :
Programming the Pin Map

Programming the Pin Map
A pin map describes the device pins by assigning a pin type to each device pin name. For each RF DUT pin in a new or existing Pin Map sheet:

1. Type a name for the device pin, such as RF_in, in the Pin Name column.
2. Select Analog as the pin type, in the Type column.

For more information, see [Pin Map Sheet](#) in the DataTool section.

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UltraWave24 Instrument Programming : IG-XL Programming for UltraWave24 Instruments : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see [User View of UltraWave24 SMPM Connector Blocks](#), [User View of UltraWave24 DIB Pad Patterns](#), and [UltraWave24 Signal Descriptions](#).

The UltraWave24 uses the following channel types:

MW	Identifies a bidirectional channel (port) on a MW Measure HD board. MW channels are compatible with UltraWave 12G.
MWHD	Identifies a bidirectional channel (port) on a MW Measure HD board that has full UltraWave24 support. MWHD channels are incompatible with UltraWave 12G.
MWSource	Identifies a DUT Clock reference pin on a MW Synthesizer HD board. MW channels are compatible with UltraWave 12G.

The MWSource channel type supports only the MWSource logical instrument. IG-XL allows you to use the MWSource channel type on a bidirectional port, but then you cannot use any other MW logical instruments. There is no advantage to using the MWSource channel type on a bidirectional port. Use the MWSource channel type with the MWSource logical instrument to program DUT Clock pins.

The UltraWave24 SynthHD board defines the channels listed below, which are a superset of the UltraWave 12G Synth board channels. These channels are generated for each type of UltraWave24 configuration. The z in the channel map names is the corresponding test head slot.

UltraWave SynthHD Board Channels

Tester Channel	Channel Map Pogo Name	Channel Map Signal Name	Sidewalk 4 Location	UltraWave 12G Compatible?
RF5	z.e1	z.RefClk1A	R414-R415	Yes
RF6	z.d2	z.RefClk1B	M411-M412	Yes
RF7	z.c1	z.RefClk1C	H414-H415	Yes
RF8	z.b2	z.RefClk1D	E411-E412	Yes
RF9	z.e4	z.RefClk1E	R405-R406	No
RF10	z.d5	z.RefClk1F	M402-M403	No
RF11	z.c4	z.RefClk1G	H405-H406	No
RF12	z.b5	z.RefClk1H	E402-E403	No
DA1	z.a3	z.RefClk1DA	B408-B409	No

The UltraWave24 Measure HD board defines the channels listed below. The x in the channel map names is the corresponding the test head slot.

UltraWave Measure HD Board Channels

Tester Channel	Channel Map Pogo Name ¹	Compatible Channel Map Signal Name ²	UltraWave24 HD Channel Map Signal Name ³
RF1	x.smpm1	x.MW1A	x.MWHD1A
RF2	x.smpm2	x.MW1B	x.MWHD1B
RF3	x.smpm3	x.MW1C	x.MWHD2A
RF4	x.smpm4	x.MW1D	x.MWHD2B
RF5	x.smpm5	x.MW2A	x.MWHD3A
RF6	x.smpm6	x.MW2B	x.MWHD3B
RF7	x.smpm7	x.MW2C	x.MWHD4A
RF8	x.smpm8	x.MW2D	x.MWHD4B
RF9	x.smpm11	x.MW3A	x.MWHD5A
RF10	x.smpm12	x.MW3B	x.MWHD5B

RF11	x.smpm13	x.MW3C	x.MWHD6A
RF12	x.smpm14	x.MW3D	x.MWHD6B
RF13	x.smpm15	x.MW4A	x.MWHD7A
RF14	x.smpm16	x.MW4B	x.MWHD7B
RF15	x.smpm17	x.MW4C	x.MWHD8A
RF16	x.smpm18	x.MW4D	x.MWHD8B
RF17	x.jib2_smpm2		x.MWHD1C
RF18	x.jib2_smpm3		x.MWHD2C
RF19	x.jib2_smpm7		x.MWHD3C
RF20	x.jib2_smpm8		x.MWHD4C
RF21	x.jib2_smpm12		x.MWHD5C
RF22	x.jib2_smpm13		x.MWHD6C
RF23	x.jib2_smpm17		x.MWHD7C
RF24	x.jib2_smpm18		x.MWHD8C

- 1 The sidewalk 1 connector block uses the same SMPM Pogo names to identify the signal location at the DIB.
- 2 These UltraWave 12G compatible signal names can be used with MW and MWSource channel types.
- 3 These UltraWave24 signal names can only be used with the MWHD channel type.

To program the MW channel type in a channel map, choose the corresponding instrument type:

- All MW Instruments Except DUT Clock
- DUT Clock

All MW Instruments Except DUT Clock

For each RF pin in a new or existing channel map, for all MW instruments except a DUT Clock:

1.

Select MW or MWHD as the tester channel type, in the Type column.

2. Enter a Pogo pin name or signal name to the Tester Channel Site column to assign a MW tester channel to the pin.

DUT Clock

For each RF pin in a new or existing channel map for a DUT Clock:

1. Select MWSource as the tester channel type, in the Type column.
2. Enter a Pogo pin name or signal name to the Tester Channel Site column to assign a MW tester channel to the pin.

The figures below show a channel map for source, measure, and DUT clock pins using the Signal and Pogo names. All of these pins are compatible with the UltraWave 12G since they do not use the MWHD channel type.

Channel Map

TERAD

DIB ID:

View Mode: Signal

Site Replication:

Device Under Test		Tester Channel				
Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
rf_source		MW	4.MWHD7A	4.MWHD1A	2.MWHD7A	2.MWHD1A
rf_measure		MW	4.MWHD6A	4.MWHD4A	2.MWHD6A	2.MWHD4A
ref_clk		MWSource	10.RefClk1D	10.RefClk1C	10.RefClk1B	10.RefClk1A

UltraWave24 Channel Map Using Signal Names

Channel Map

TERAD

DIB ID:

View Mode: Pogo

Site Replication:

Device Under Test		Tester Channel				
Pin Name	Package Pin	Type	Site 0	Site 1	Site 2	Site 3
rf_source		MW	4.smpm15	4.smpm1	2.smpm15	2.smpm1
rf_measure		MW	4.smpm13	4.smpm7	2.smpm13	2.smpm7
ref_clk		MWSource	10.b2	10.c1	10.d2	10.e1

UltraWave24 Channel Map Using Pogo Names

For more information, see Channel Map Sheet in the DataTool section.

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UltraWave24 Instrument Programming : IG-XL Programming for UltraWave24 Instruments : DIB Access

DIB Access

UltraWave24 RefClk channels on the Synth board support DIB access.

Note, however, that the UltraWave24 Synth board has only one DIB input multiplexed across the eight RefClk outputs. Attempts to connect through DIB access may connect to the wrong RefClk channel on the same board that already has a DIB access connection.

The last applied DIB access connection takes precedence.

For more information, see [DIB Access](#).

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UltraWave24 Instrument Programming : Using UltraWave24 in Concurrent Testing

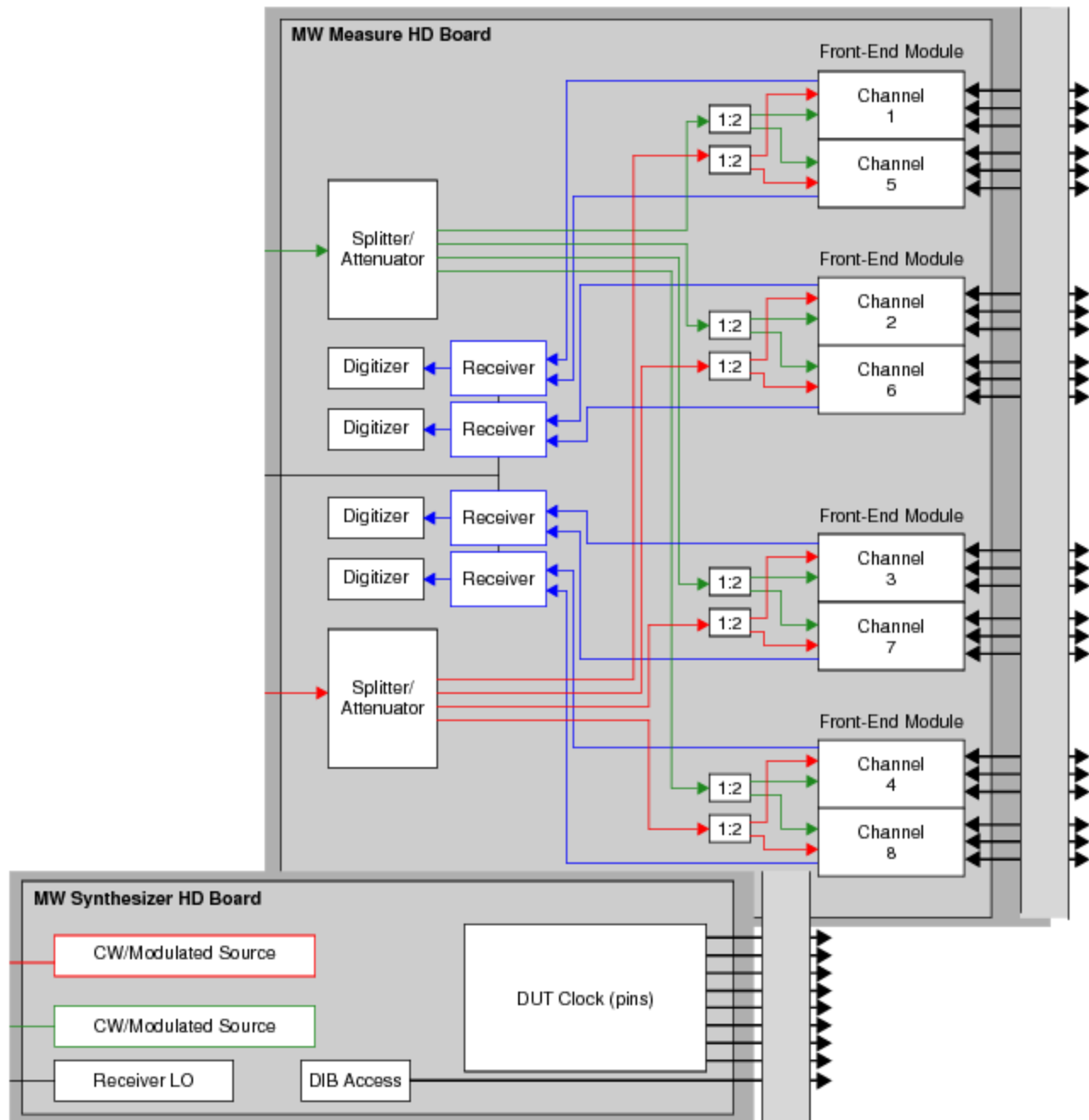
Using UltraWave24 in Concurrent Testing

To support concurrent testing, you can assign each UltraWave hardware block to unique time domains. Concurrent features for UltraWave include:

- A MW Measure HD board can belong to up to four distinct time domains. This provides support for up to 16 time domains on four MW Measure HD boards.
- A MW Synthesizer HD board can belong to up to two distinct time domains
- Each UltraWave hardware block can be controlled by unique time domains simultaneously using Sync-Link. Features include:
 - Independent access to Sync-Link
 - Independent sample clock programming
 - Independent clock synchronization with other UltraWave hardware blocks or with digital clocks
 - Independent control by digital pattern
 - Independent capture memory and triggers

- Multiple hardware blocks can be grouped into a common time domain
- UltraWave24 fully supports flow domains and asynchronous pattern starts

The following diagram shows the granularity of time domain programming for UltraWave boards and logical instruments. Each white box in the diagram is a hardware block. For example, each Front End Module on a Measure board can be assigned to distinct time domains (up to four per board).



UltraWave24 Time Domain Granularity

For more information, see [About Concurrent Test](#).

[Duplex MTD Example](#)

The following example shows the elements that are included in an UltraWave test program for multiple time domains. This example describes a program that uses an MWSource in one time domain for the RX side of a device and an MWReceiver in another time domain for the TX side of the device.

The following figures show an example pin map and channel map.

Pin Map

Group Name	Pin Name	Type
	Source	Analog
	Measure	Analog
	Dig1	I/O
	Dig2	I/O
MWSourceOnly	Dig1	TimeDomain
MWSourceOnly	Source	
MWReceiveOnly	Dig2	TimeDomain
MWReceiveOnly	Measure	

Example Pin Map

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
Source		MW	2.MW1A	2.MW2A
Measure		MW	2.MW3A	2.MW4A
Dig1		I/O	1.ch16	1.ch17
Dig2		I/O	1.ch128	1.ch129

Example Channel Map

For most UltraWave24 instruments, mapping of the front end module used for each channel is not complex since the module is the channel map number. So, in this example:

- Pin Source uses front end module 1 (MW1A) for Site 0 and module 2 (MW2A) for Site 1.

- Pin Measure uses front end module 3 (MW3A) for Site 0 and module 4 (MW4A) for Site 1.

However, mapping of MWSource and MWReceiver is not as obvious. This figure below shows the best way to partition resources, mapping odd sites to the even front end modules and even sites to the odd modules. This organization expands in the same way for more sites and measure boards.

MW Channel	Site 0				Site 1			
	1		3		2		4	
	Src X	Rcv X	Src Y	Rcv X	Src X	Rcv Y	Src Y	Rcv Y
A								Concurrency Group 1
B								
C								Concurrency Group 2
D								

Mapping of MWSource and MWReceiver for Concurrent Test

You must also include VBT code to OnProgramLoaded:

```
TheHdw.MW.EnableIntraSlotMTD = True
```

So, the function would look something like this:

```
Function OnProgramLoaded()
```

```
    On Error GoTo errHandler
```

```
    ' Put code here
```

```
    TheHdw.MW.EnableIntraSlotMTD = True
```

```
Exit Function
```

errHandler:

```
    HandleExecIPError "OnProgramLoaded"
```

```
End Function
```

For more information, see [EnableIntraSlotMTD](#) in the VBT Syntax Reference.

If you want to control pattern interlocking while a pattern is running, add the following to OnProgramStarted:

```
TheHdw.MW.POPInterlock = tIMWPOPIterlockEnable
```

So, the function would look something like this:

```
Function OnProgramStarted()
```

```
    On Error GoTo errHandler
```

```
    ' Put code here
```

```
    TheHdw.MW.POPInterlock = tIMWPOPIterlockEnable
```

```
Exit Function
```

errHandler:

```
    HandleExecIPError "OnProgramStarted"
```

```
End Function
```

For more information, see [POPInterlock](#) in the VBT Syntax Reference.

The program also includes patterns defined as follows:

- One time domain would contain tester source instruments (such as ModSrc) for the RX side of the DUT

- One time domain would contain tester receive instruments (such as MWReceiver) for the TX side of the DUT

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UltraWave24 Instrument Programming : UltraWave24 Pattern Control

UltraWave24 Pattern Control

You can control an UltraWave24 logical instruments using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor.

Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as the pattern microcode used for controlling the instrument during pattern execution.

This section includes general topics on UltraWave24 pattern control. The following topics are available:

- Using UltraWave24 Instruments with POP

- Understanding UltraWave Pattern Interlocking

For more information on pattern control for UltraWave24 instruments, see:

- MWModulatedSource Pattern Control

- MWMultiToneSource Pattern Control

- MWNoiseSource Pattern Control

- MWReceiver Pattern Control

•	MWSource Pattern Control
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•	UltraWave24 VNA Pattern Control
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UltraWave24 Instrument Programming : UltraWave24 Pattern Control : Using UltraWave24 Instruments with POP

Using UltraWave24 Instruments with POP

Pattern oriented programming (POP) is a test programming style that has many benefits including measurement stability, test reliability, and test time. This type of programming can be used with UltraWave logical instruments.

POP uses pattern memory and Digital instrument boards that are present on all UltraFLEX testers. The PatGens on the Digital instruments control the application of patterns and can respond to DUT and instrument conditions in real time. The PatGens also maintain synchronization with each other through the SyncLink. Pattern memory contains not only digital 1s and 0s but also analog instrument opcodes. The analog instrument opcodes perform functions like applying a PSet, performing a strobe, or starting a signal using Start.

Because of this test system architecture, once a pattern is started, the instruments can handle all aspects of a test: applying conditions, performing measurements, and reacting to changing test conditions. And because timing and synchronization are handled by the hardware, test repeatability and precision is superior to what the test computer can achieve.

While a pattern is the executing a test, the test computer can communicate with tester resources through the test system bus or can process data.

When using an UltraWave logical instrument (MWModulatedSource or other instrument), any vector that contains a POP microcode (either LoadSettingsConnect or Disconnect) must be fully defined across all defined logical instrument instances. IG-XL returns an error if you do not follow this guideline.

Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

The following figure shows an example pattern file (UltraWavePOP.PAT) in PatternTool that uses POP microcodes and several UltraWave logical Instruments. Note the following:

- For vectors with POP microcodes, the example pattern demonstrates the idea of defining the state for all logical instruments. Vectors 0, 7, 14, and 20 contain POP microcodes.
- Vector 14 includes a comment that this vector will not cause a glitch on the MWSource pin (as the same Signal is reapplied). This same concept applies to all sourcing logical instruments any time the same Signal is reapplied.

UltrawavePOP.PAT (vm_UltraWavePOP - VM)									
TSet	Command	Label	Vector	Source (MWModulatedSource)	Source (MWMultitoneSource)	Measure (MWReceiver)	Source (MWSource)	Dig1	Comment
Basic	start_label UltrawavePOP		0	LoadSettingsConnect GSM	Disconnect	LoadSettingsConnect GSM	Disconnect	0	
Basic	repeat 10000		1					0	1mS wait time
Basic			2	Start GSM				0	
Basic	repeat 100		3					0	10uS wait time
Basic			4			Trig GSM		0	
Basic	repeat 10000		5					0	1mS wait time
Basic			6					0	
Basic			7	Disconnect	Disconnect	LoadSettingsConnect WCDMA	LoadSettingsConnect WCDMA	0	
Basic	repeat 10000		8					0	1mS wait time
Basic			9					0	
Basic	repeat 100		10					0	10uS wait time
Basic			11			Trig WCDMA		0	
Basic	repeat 10000		12					0	1mS wait time
Basic			13					0	
Basic			14	Disconnect	Disconnect	LoadSettingsConnect WCDMA2	LoadSettingsConnect WCDMA2	0	No glitch created on MWSource, same
Basic	repeat 10000		15					0	1mS wait time
Basic			16					0	
Basic	repeat 100		17					0	10uS wait time
Basic			18			Trig WCDMA2		0	
Basic	repeat 10000		19					0	1mS wait time
Basic			20	Disconnect	Disconnect	Disconnect	Disconnect	0	
Basic	repeat 2000		21					0	200uS wait time
Basic			22					0	
Basic			23					0	

Vertical: 65 Vectors

Vectors 0 to 23

POP Example Pattern Using UltraWave Instruments

The following is the ASCII code for the example pattern.

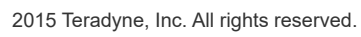
```
instruments = {
    Source:MWModulatedSource 1;
    Source:MWMultitoneSource 1;
    Measure:MWReceiver 1;
    Source:MWSource 1;
}

vm_vector
vm_UltraWavePOP          ( $tset, Dig1)
{
start_label Ultrawave_POP:
(Source:MWModulatedSource = LoadSettingsConnect GSM)
(Source:MWMultitoneSource = Disconnect)
```

```
(Measure:MWRceiver = LoadSettingsConnect GSM)
(Source:MWSource = Disconnect)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
(Source:MWModulatedSource = Start GSM)
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig GSM)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
(Source:MWModulatedSource = Disconnect)
(Source:MWMultitoneSource = Disconnect)
(Measure:MWRceiver = LoadSettingsConnect WCDMA)
(Source:MWSource = LoadSettingsConnect WCDMA)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig WCDMA)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
(Source:MWModulatedSource = Disconnect)
(Source:MWMultitoneSource = Disconnect)
(Measure:MWRceiver = LoadSettingsConnect WCDMA2)
(Source:MWSource = LoadSettingsConnect WCDMA)
    > Basic 0 ; //No glitch created on
    //MWSource because same
    //Signal (WCDMA) is
    //reapplied.
repeat 10000    > Basic 0 ; //1ms wait time
    > Basic 0 ;
repeat 100    > Basic 0 ; //10us wait time
(Measure:MWRceiver = Trig WCDMA2)
    > Basic 0 ;
repeat 10000    > Basic 0 ; //1ms wait time
(Source:MWModulatedSource = Disconnect)
```

```
> Basic 0 ;
```

```
halt
}
```



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UltraWave24 Instrument Programming : UltraWave24 Pattern Control : Understanding UltraWave Pattern Interlocking

Understanding UltraWave Pattern Interlocking

By default, hardware for UltraWave24 instruments is set to listen to programming commands that are initiated from VBT and that originate from the host computer. When a digital pattern is burst that contains UltraWave microcodes, such as LoadSettingsConnect, the hardware is activated to also listen for instructions that originate from the digital pattern generator which are sent through the Instrument Sync Link (ISL) bus.

While an UltraWave instrument is set to listen for ISL instructions, IG-XL locks out all commands originating from VBT, and a run-time error is generated if you attempt to program the UltraWave instrument from VBT while the pattern is running. This pattern interlocking is done to avoid collisions in the hardware in the case where instructions from different sources occur too close together for the hardware to react properly to both instructions.

The lockout starts when a digital pattern containing LoadSettingsConnect or Disconnect microcodes is started. The lockout is released at end of a Digital.Patgen.Haltwait VBT statement indicating that the pattern burst is complete or when an explicit Patgen.Halt method is executed. At times, however, a test program may be written such that VBT commands are to be issued to an UltraWave instrument while a pattern is running. For example, you might need to issue commands while a pattern is looping on a wait for CPU flag. In such cases, you can override the pattern lockout to allow VBT programming to occur. The syntax is:

```
TheHdw.MW.PatternMicroCodeControl = tlMWPatternMicroCodeControlOff
```

This lockout is controlled per time domain for UltraWave instruments.

Once the needed VBT commands are issued, restore .PatternMicroCodeControl to tlMWPatternMicroCodeControlOn before continuing pattern execution.

Note that changing the .PatternMicroCodeControl property only toggles the VBT interlock mechanism. It does not disable ISL commands. Collisions are still possible when

PatternMicroCodeControl is set to tlmwPatternMicroCodeControlOff and the digital pattern is still running and executing microcodes. Take care when toggling .PatternMicroCodeControl such that this does not occur.

For more information, see [PatternMicroCodeControl](#) in the VBT Syntax Reference.

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UltraWave24 Theory of Operation

UltraWave24 Theory of Operation

The UltraWave24 subsystem provides microwave instrument hardware functions on boards and splitter modules in the test head. The hardware functions are allocated, configured, and connected in different ways to provide MW instrument capabilities. These software configured MW instruments are called logical instruments. The operation of individual logical instruments depends on the operation of the subsystem functions.

This following topics are available:

- [UltraWave24 Subsystem Architecture](#)
- [UltraWave24 Hardware Functions](#)
- [UltraWave24 Logical Instruments and Hardware Functions](#)
- [User View of UltraWave24 SMPM Connector Blocks](#)
- [User View of UltraWave24 DIB Pad Patterns](#)
- [UltraWave24 Signal Descriptions](#)

For information on UltraWave24 instrument performance specifications and standards, see [UltraWave24 Specifications](#).

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UltraWave24 Theory of Operation : UltraWave24 Subsystem Architecture

UltraWave24 Subsystem Architecture

All hardware functions in the UltraWave24 subsystem are on boards and splitter modules in the test head. The figure shows the basic, high-level hardware functions and signal connections for a typical configuration having 24 RF ports: one MW Measure HD board and one MW Synthesizer HD board. An MW Synthesizer HD board includes:

- An LO synthesizer for the down-converting receivers on one or more MW Measure HD boards
- Two synthesizers and modulators for use as a CW or modulated signal generator for the DUT
- A DUT clock (through spring-loaded pins)

A MW Measure HD board includes:

- A single LO source input for all receivers, going into a 1:4 splitter
- Four receivers, each connected to a dedicated digitizer
- Two signal source inputs, each going into an attenuator/1:4 splitter
- Four front-end modules, each of which has two channels. Each channel includes:
 - A choice of two source signal inputs from an attenuator/1:4 splitter

- [Receive signal output](#)

- [1:3 bidirectional mux to RF DIB ports](#)

- [Two optional IF ports that allow access to the downconverted signal from the two microwave receivers, with 500+ MHz of IF bandwidth](#)

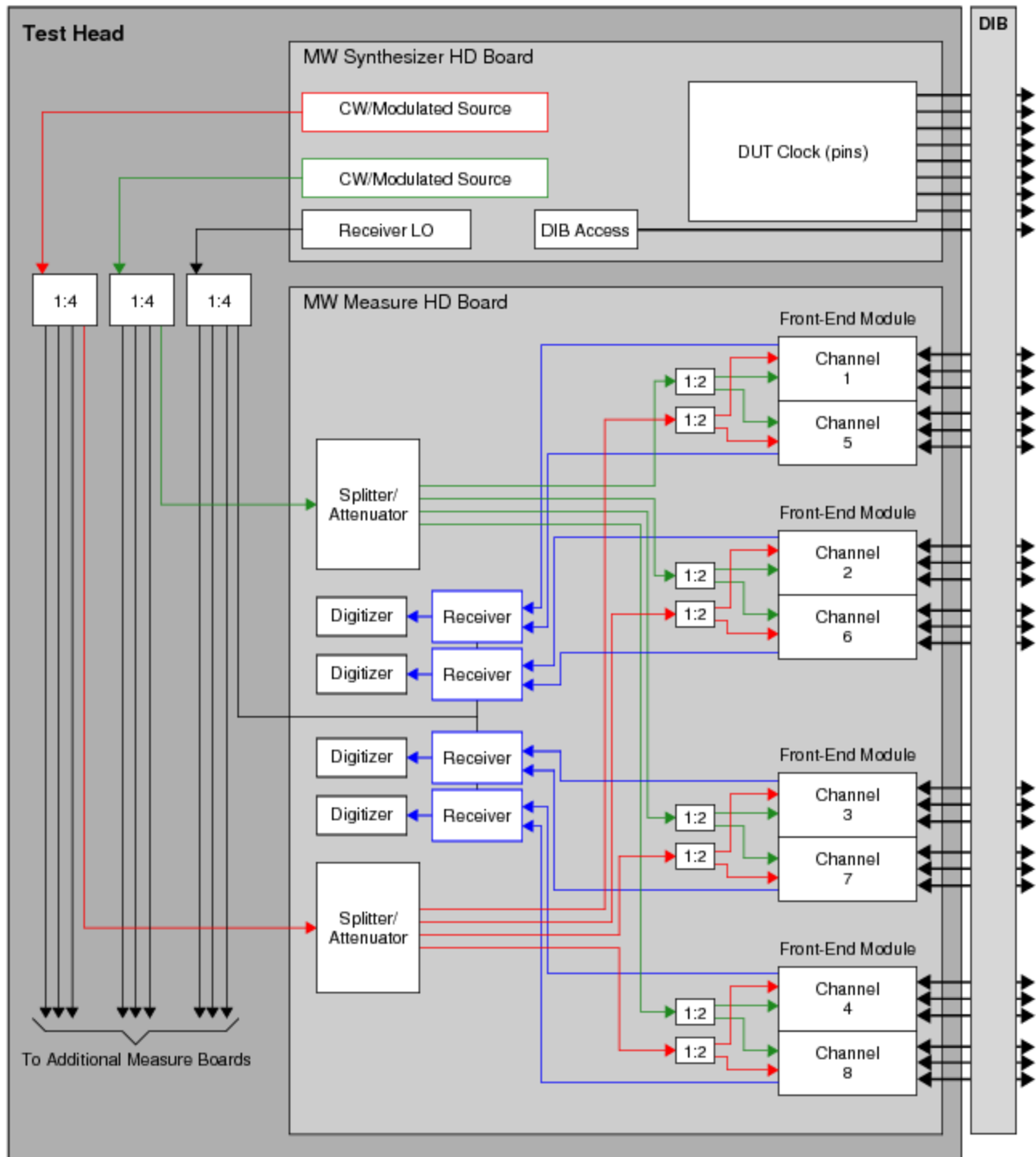
The UltraWave24 subsystem is expandable for high pin-count devices and multiple sites. The MW Synthesizer HD board is connected to 1:4 splitter modules in the test head which can in turn support up to four measure boards (96 RF ports). For more information, see [Supported UltraWave24 Configurations](#).

For more information on UltraWave24 architecture, see the following topics:

- [Architecture and Resource Mapping](#)

- [Shared Hardware Functions](#)

- [Supported UltraWave24 Configurations](#)



UltraWave24 Subsystem Hardware Architecture

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UltraWave24 Theory of Operation : UltraWave24 Subsystem Architecture : Architecture and Resource Mapping

Architecture and Resource Mapping

The UltraWave24 has a fanout architecture that allows increasing port count by adding MW Measure HD boards. The fanout ratio is 1:24 when a MW Synthesizer HD board is connected to one measure board and 1:96 when two MW Synthesizer HD boards are connected to four measure boards.

UltraWave24 has a single LO source, so that the LO is common to all receivers. Each measure board has four receivers, each with a digitizer. Each digitizer can connect to 1/4 of the total ports (up to six ports) per measure board.

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UltraWave24 Theory of Operation : UltraWave24 Subsystem Architecture : Shared Hardware Functions

Shared Hardware Functions

The UltraWave24 subsystem architecture requires MW logical instruments to share certain hardware functions:

- Each measure board (up to four) shares the synthesizers for the receiver LO and CW/modulator generator functions with the other measure boards through 1:4 splitters. Because the synthesizers are shared:
- All active receiver ports must use the same frequency settings.
- All active source ports using an individual CW/modulated source must use the same frequency settings and CW/modulated signal content. Fine attenuation level settings can be different.
- Each front-end module on a measure board shares a synthesizer for either of the two CW/modulated sources on the synthesizer board with another front-end module through a 1:4 splitter with leveling (see pairs on the diagram).

Because synthesizers are shared, all active source ports on the pair of front-end modules must use the same frequency settings and CW/modulated signal content.

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UltraWave24 Theory of Operation : UltraWave24 Subsystem Architecture : Supported UltraWave24 Configurations

Supported UltraWave24 Configurations

The capabilities of the supported UltraWave24 configurations depend on the number of MW Measure HD boards. This number determines the number of ports, up to the DIB cabling limit. The following table lists the supported configurations.

Supported UltraWave24 Configurations		
Synthesizer Boards	Independent Synthesizers	RF Ports
1 MW Synthesizer HD	1 Receiver LO 2 CW/Modulated Sources (ex: 1, 2) 1 DUT reference	1 Measure Board = 24 ports Eight Source paths, each of which can choose between 1 or 2 2 Measure Boards = 48 ports Twice the capacity, but the same source selection, as the 1x board.

2 MW Synthesizer HD

- 2x Receiver LO
- 4x CW/Modulated Sources
(ex: A1, A2, B1, B2)
- 2x DUT reference

2 Measure Boards = 48 ports
Two groups of eight source paths
(one on each board):

- One board's eight source paths can each choose between A1 and A2.
- The other board's eight source paths can each choose between B1 and B2.

4 Measure Boards = 96 ports
Twice the capacity, but the same source selection, as the 2x board

A minimum subsystem configuration (24 RF ports) has two boards:

- | |
|-------------------------------|
| • One MW Synthesizer HD board |
|-------------------------------|

- | |
|---------------------------|
| • One MW Measure HD board |
|---------------------------|

In this configuration, two CW/modulated sources are available. Both sources are connected to the 1:4 splitter on the measure board. This allows a choice of either source at all ports.

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[UltraWave24 Theory of Operation](#) : UltraWave24 Hardware Functions

UltraWave24 Hardware Functions

The UltraWave24 subsystem can be represented by hardware function blocks that support the MW logical instruments. The logical instruments connect the functions in a number of ways for specific source and measure capabilities. The functions and connections are described in the following topics:

- | | |
|---|--|
| • | UltraWave24 Subsystem Function Block Diagram |
| • | UltraWave24 Function Descriptions |

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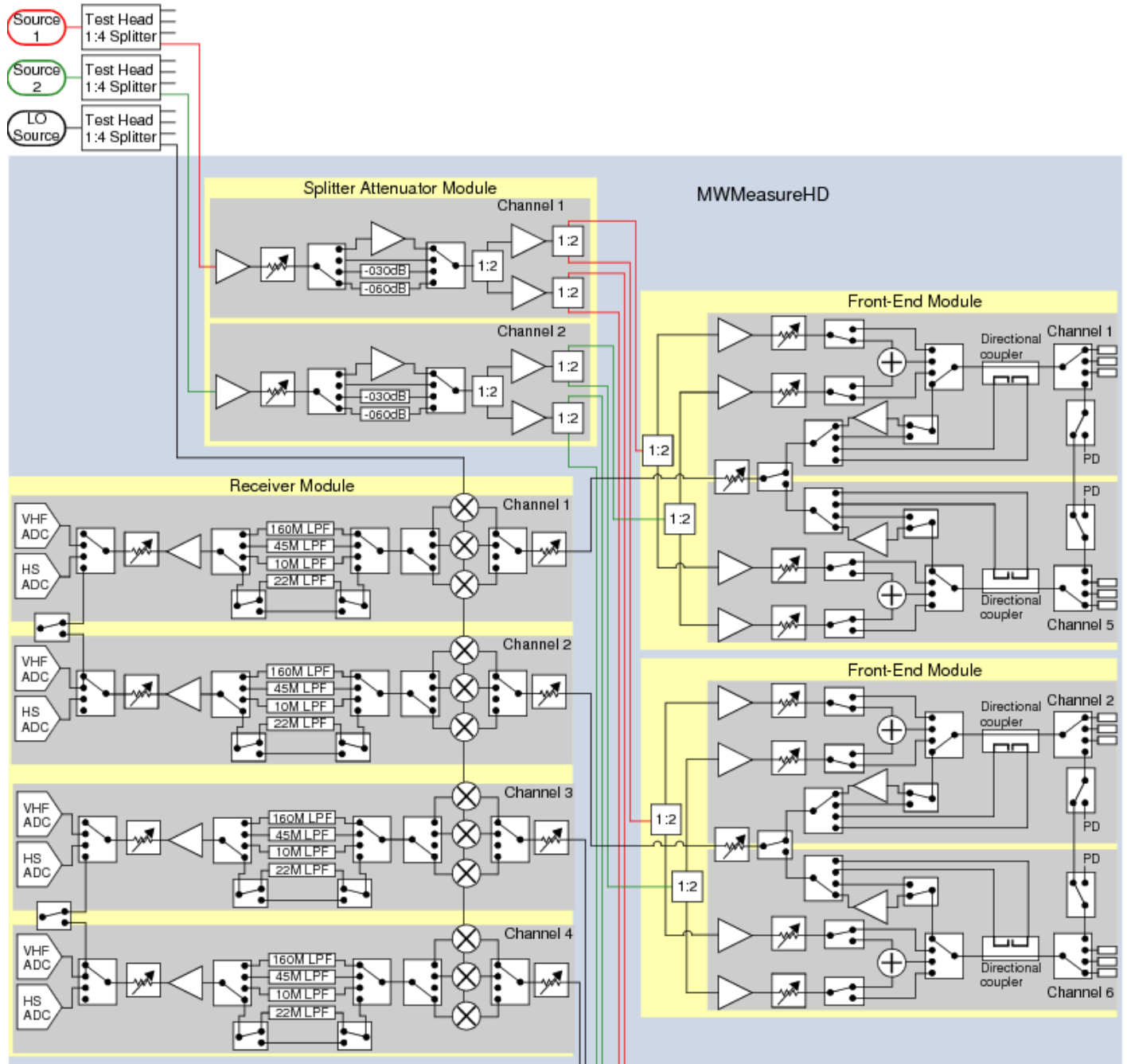
[UltraWave24 Theory of Operation](#) : [UltraWave24 Hardware Functions](#) : [UltraWave24 Subsystem Function Block Diagram](#)

UltraWave24 Subsystem Function Block Diagram

The figure shows UltraWave24 function blocks and switch functions that connect MW functions in various ways to create and control logical instruments. The switch arrangements represent how the functions can be logically connected and how signals flow, but are not necessarily identical to physical hardware switches. Any MW logical instrument, and any valid combination of logical instruments operating simultaneously, can be represented as a group of functions and switch settings on the diagram.

The block diagram shows an example system with a MW Synthesizer HD board and one MW Measure HD board. All DIB connections are RF ports, except the connections for the DUT clock. For more information on the function blocks, see [UltraWave24 Function Descriptions](#). For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see:

- [User View of UltraWave24 SMPM Connector Blocks](#)
- [User View of UltraWave24 DIB Pad Patterns](#)
- [UltraWave24 Signal Descriptions](#)



MWMesureHD Functional Diagram, Showing Two of Four Front End Modules



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[UltraWave24 Theory of Operation](#) : [UltraWave24 Hardware Functions](#) : [UltraWave24 Function Descriptions](#)

UltraWave24 Function Descriptions

The UltraWave24 Subsystem includes the following functions:

1:2	1:2 power splitter. Splits the following signals: - Receiver LO for two receivers on a measure board - Attenuated source from CW/Modulated synthesizer for two source channels on a measure board - Receiver I/F output between the digitizer and DIB I/F output
1:4	1:4 power splitter. Splits the following signals: - Receiver LO output for one to four measure boards - CW/Modulated synthesizer output for sources on one to four measure boards
Bidirectional Coupler 	RF signal coupler for incident waves. Includes incident signal coupler and reflected signal coupler. Required for s-parameter measurements.

CW/Modulated Synthesizer	Synthesizer for CW (continuous wave or sine) signals and modulated signals.
DUT Clock	Reference clock for a DUT or DIB circuit. Available to the DIB at four Pogo (spring-loaded) pins.
Fine Attenuator	Attenuator with very fine steps. Provides fine adjustments to adjust source signals to specified level accuracy. Can also adjust levels at ports to different levels.
HS ADC	High speed analog-to-digital converter (500kHz-45MHz BW, 16 bit).
IF output	IF output of the receiver down-converter. Switched and available to the DIB at an RF (subminiature coax) port.
I/Q Mod	I/Q modulator for the modulated source.
LNA 	Low-noise amplifier in receive signal path. Bypassed when not needed.
PD	Power detector.
Pogo pin	High frequency tester pin. Spring-load tester pin on sidewalk 4 at the DIB interface. Contrast to RF port which is a subminiature coax connector.
Receiver	RF receiver (mixer/downconverter).
Receiver LO Synthesizer	Synthesizer of LO signal for receivers.
RF Port	Bidirectional RF connection at the DIB interfaced. Multiplexed 1:4 for source and receive channels.
Signal Combiner 	Adds (combines) two RF signals. Allows two sources to create a multitone signal.
Splitter Attenuator	Attenuator with large (30dB) steps. Provides approximate level to each source channel. Exact level is set by the fine and step attenuators.
VHF ADC	VHF analog-to-digital converter (5MHz-165MHz BW, 12 bit).

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UltraWave24 Theory of Operation : UltraWave24 Logical Instruments and Hardware Functions

UltraWave24 Logical Instruments and Hardware Functions

Any UltraWave24 logical instrument or any valid combination of instruments operating simultaneously can be represented as a group of functions and switch settings on the UltraWave24 Subsystem Function Block Diagram.

The following topics describe the configuration of the UltraWave24 subsystem for the operation of each logical instrument:

- [UltraWave24 MWModulatedSource Theory of Operation](#)
- [UltraWave24 MWMultiToneSource Theory of Operation](#)
- [UltraWave24 MWNoiseSource Theory of Operation](#)
- [UltraWave24 MWReceiver Theory of Operation](#)
- [UltraWave24 MWSource Theory of Operation](#)
- [UltraWave24 VNA Theory Of Operation](#)

The topics also provide information on hardware functions during operation of the specific logical instrument.

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UltraWave24 Theory of Operation : User View of UltraWave24 SMPM Connector Blocks

User View of UltraWave24 SMPM Connector Blocks

This section shows the UltraWave24 subsystem SMPM connector locations as seen from the user perspective (looking at the top, or device side, of the DIB/HIB/PIB and toward the test system). Two boards in the UltraWave24 subsystem have SMPM connectors on sidewalk 1:

- | | |
|---|----------------------------------|
| • | UltraWave24 Measure Board, JIB 1 |
| • | UltraWave24 Measure Board, JIB 2 |

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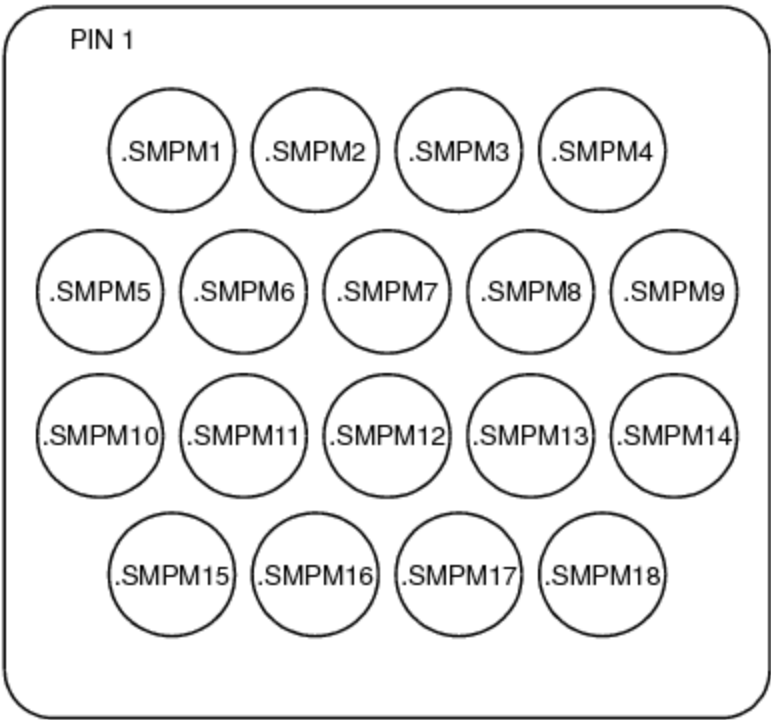
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UltraWave24 Theory of Operation : User View of UltraWave24 SMPM Connector Blocks :
UltraWave24 Measure Board, JIB 1

UltraWave24 Measure Board, JIB 1

User View



User View

User View

UltraWave24 Measure Board SMPM Connectors on Sidewalk 1 (User View)

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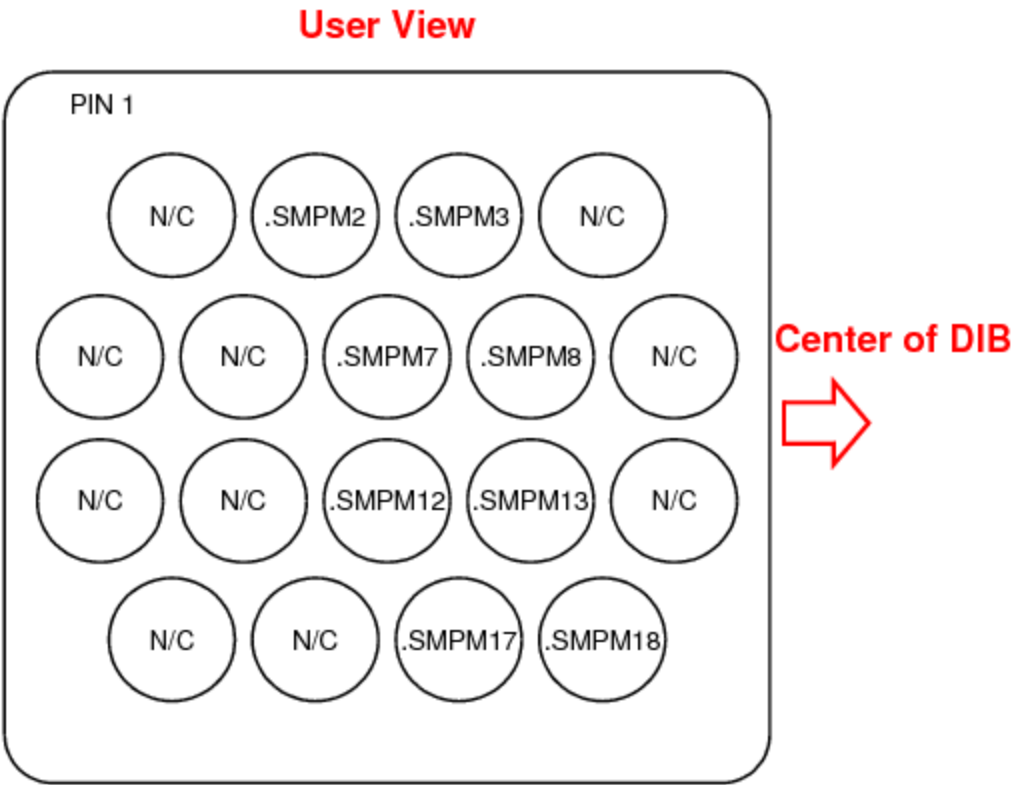
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UltraWave24 Theory of Operation : User View of UltraWave24 SMPM Connector Blocks :
UltraWave24 Measure Board, JIB 2

UltraWave24 Measure Board, JIB 2



User View

User View

UltraWave24 Synthesizer Board SMPM Connectors on Sidewalk 1 (User View)
For Reference (DUT) Clock connectors, see User View of UltraWave24 DIB Pad Patterns.

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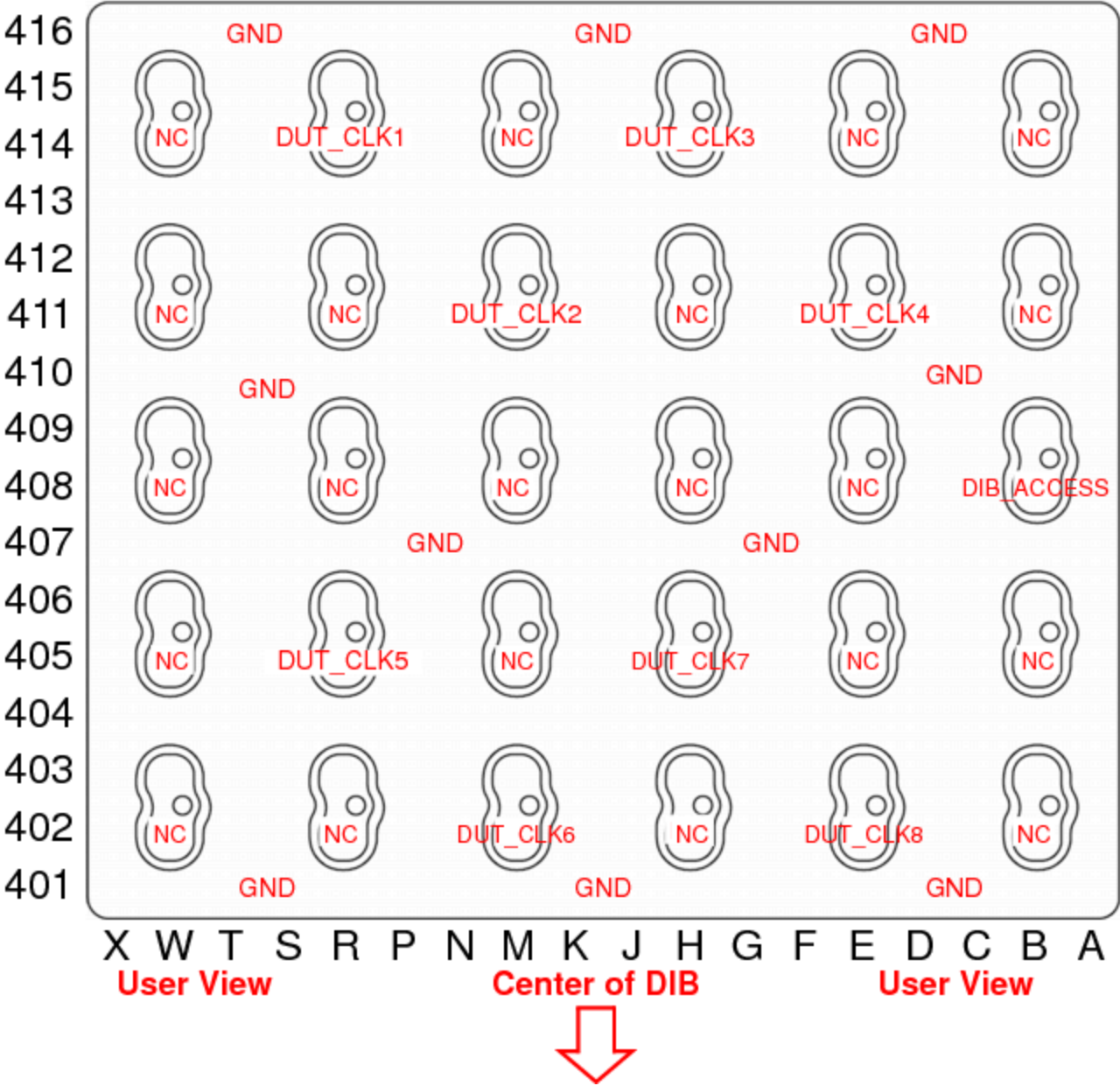
	
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[UltraWave24 Theory of Operation](#) : User View of UltraWave24 DIB Pad Patterns

User View of UltraWave24 DIB Pad Patterns

This section shows the UltraWave24 subsystem DIB pad connector locations as seen from the user perspective (looking at the top, or device side, of the DIB/HIB/PIB and toward the test system). The MW Synthesizer HD board is the only board in the UltraWave24 subsystem that has DIB pad connectors on sidewalk 4.

UltraWave24 Synthesizer Reference Board, Ref Clock



UltraWave Synthesizer Reference Clock Pad Pattern for Sidewalk 4 (User View)

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[UltraWave24 Theory of Operation](#) : UltraWave24 Signal Descriptions

UltraWave24 Signal Descriptions

For information on UltraWave24 signals, see [UltraWave24 DIB Design Guidelines](#).

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About the UltraWave24 Subsystem

About the UltraWave24 Subsystem

The UltraWave24 subsystem provides a test instrument for sourcing, measuring, and analyzing RF signals in the frequency range from 50 MHz to 6 GHz. The UltraWave24 subsystem is made up of instrument function modules on boards and modules in the test head, and instrument software. IG-XL software provides MW logical instruments that use specific combinations of function modules to source and capture signals for RF measurements. The logical instrument interfaces simplify programming of hardware resources.

Testing applications include mobile phone transceivers, wireless LAN chipsets, RF amplifiers, TV tuners, and IF devices. Typical tests are RF power, gain, IP3, noise figure, and phase noise. This section includes the following topics:

- | | |
|---|---|
| • | Features |
| • | Safety Information |
| • | Key Terms |
| • | UltraWave24 Software Licensing Requirements |

UltraWave24 subsystem information includes the following sections:

- | | |
|---|--|
| • | UltraWave24 Theory of Operation |
| • | UltraWave24 Instrument Calibration |

- [UltraWave24 Instrument Programming](#)
- [UltraWave24 Displays](#)
- [UltraWave24 Examples](#)

Related Topics

- [UltraWave24 Specifications](#)
- [UltraWave24 DIB Design Guidelines](#)
- The UltraWave24 subsystem supports all UltraWave24 MW logical instruments. For more information, see:
 - [About the UltraWave24 MWModulatedSource](#)
 - [About the UltraWave24 MWMultiToneSource](#)
 - [About the UltraWave24 MWNoiseSource](#)
 - [About the UltraWave24 MWReceiver](#)
 - [About the UltraWave24 MWSource](#)
 - [About the UltraWave24 VNA](#)

